Mini project report on

Blood Bank Management and Donor-Recipient Matching System

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering

UE23CS351A – DBMS Project

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Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

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CERTIFICATE

This is to certify that the mini project entitled

Blood Bank Management and Donor-Recipient Matching System

is a bonafide work carried out by

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In partial fulfilment for the completion of fifth semester DBMS Project (UE24CS351A) in the Program of Study - Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period AUG. 2026 – DEC. 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 5th semester academic requirements in respect of project work.

Signature
Dr. Shilpa S

Assistant Professor

DECLARATION

We hereby declare that the DBMS Project entitled **Disasters Data Management System** has been carried out by us under the guidance of **Dr. Shilpa S, Assistant Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester AUG – DEC 2026.

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ABSTRACT

Blood banks need to track donors, stock, hospital requests and expiry dates accurately, because errors or delays can cost lives. When this is done on paper or in separate spreadsheets, units go unnoticed until they expire, donors are called back too early, and matching a request to a compatible unit takes too long. This project presents the Blood Bank Coordination System, a web-based database application that manages the whole cycle in one place, from donor registration and donation to inventory, hospital requests, matching and fulfilment.

The system uses a hybrid database design. MySQL 8 holds the core relational data in a normalized schema, along with views, triggers and the stored procedures sp_RegisterDonation and sp_ConfirmMatch. These procedures keep donation registration and match confirmation consistent. Before a donation is accepted, the system checks the donor's latest eligibility record and enforces the minimum 56-day gap between donations. Requests are matched against stock using ABO and Rh compatibility rules. A match moves through proposal, confirmation and fulfilment, and cancelling a request releases the units reserved for it. Full-text search over eligibility remarks uses MATCH ... AGAINST. MongoDB stores an append-only audit trail of every change in the audit_log collection, so the history of the system cannot be overwritten. The application still works when MongoDB is unavailable.

The front end is built with Next.js and React. It gives staff an overview of live stock, critical requests, expiring units and donors ready to donate again. Separate screens cover inventory, requests and matching, donors, camps, hospitals, users, donations, eligibility and the audit trail. User passwords are stored as bcrypt hashes. Together, the SQL and NoSQL layers and the web interface give blood bank staff a reliable and traceable way to reduce wastage, respond faster to critical requests and make sure donors and recipients are safely matched.

Keywords: Blood Bank, MySQL, MongoDB, Stored Procedures, Triggers, Next.js, Audit Trail

I wrote this from the README alone. Once I have schema.sql, I'll check it against your real tables and procedures and add specifics, such as the number of tables, if you want them.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
1.	INTRODUCTION	11
2.	PROBLEM DEFINITION	12
3.	ER MODEL	13
4.	ER TO RELATIONAL MAPPING	14
5.	DDL STATEMENTS	15
6.	DML STATEMENTS	20
7.	QUERIES (SIMPLE QUERY AND UPDATE AND DELETE OPERATION, CORRELATED QUERY AND NESTED QUERY)	23
8.	STORED PROCEDURE, FUNCTIONS AND TRIGGERS	30
9.	FRONT END DEVELOPMENT	40
REFERENCES/BIBLIOGRAPHY		41
APPENDIX A DEFINITIONS, ACRONYMS AND ABBREVIATIONS		42

LIST OF TABLES

Table No.	Title	Page No.
6.3	Insertion of records into the Eligibility table	**35**

LIST OF FIGURES

Figure No.	Title	Page No.
6.1	Insertion of records into the Hospital table	**34**
6.2	Insertion of records into the Donor table	**35**
6.4	Insertion of records into the Donation table	**36**
6.5	Insertion of records into the Patient table	**36**
6.6	Insertion of records into the Blood_Request table	**37**
6.7	Insertion of records into the Blood_Unit table	**37**

1. INTRODUCTION

1.1 OVERVIEW

Blood is a life-saving resource that cannot be manufactured and has a short shelf life, so a blood bank must always know how much blood it holds, which donors can give again, and which units are close to expiry. When this information is kept on paper or in separate spreadsheets, stock goes unnoticed until it expires, donors are contacted too early or too late, and finding a compatible donor for an urgent request takes valuable time.

This project, the Blood Bank Management and Donor-Recipient Matching System, is a relational database built in MySQL that brings these activities into one place. It records hospitals, donors, their eligibility checks, donation camps, donations, the blood units produced from each donation, patients, and the blood requests raised on their behalf. It also keeps a hospital-wise inventory and records transfers of units between hospitals.

1.2 OBJECTIVES

(i) To design a normalized relational schema covering donors, donations, blood units, inventory, patients, requests, matches and inter-hospital transfers, with primary and foreign keys enforcing integrity.

(ii) To check donor eligibility, so that only eligible donors are considered for a request.

(iii) To match each blood request with compatible donors using ABO and Rh compatibility rules, through the Compatible_Donors view and the Matching table.

(iv) To identify the best donor for a request using the stored procedure FindBestDonor, which prefers donors who have never donated or donated the longest time ago.

(v) To keep hospital inventory accurate automatically, using the UpdateInventory trigger when a unit moves between Available and Reserved, and a second trigger that marks units Expired once their expiry date has passed.

(vi) To answer operational questions such as emergency requests, low stock, expired units and camp-wise donations through SQL queries.

1.3 SCOPE OF THE REPORT

The remaining chapters present the problem definition, the ER model, the mapping of the ER model to relational tables, the DDL and DML statements with screenshots of their execution, the queries (including update and delete operations), and the stored procedure and triggers used in the system.

2. PROBLEM DEFINITION

2.1 PROBLEM STATEMENT

To design and implement a database system for a blood bank network that manages donors, donations, blood stock and hospital requests, and that can automatically suggest compatible donors for a request and keep stock levels up to date without manual bookkeeping.

2.2 LIMITATIONS OF THE EXISTING APPROACH

Manual and spreadsheet-based record keeping has several weaknesses. Stock levels are updated by hand and quickly become inaccurate. Expired units are noticed late and are wasted. There is no reliable check on whether a donor has waited long enough since the last donation or is medically deferred. Matching a patient's blood group to a donor depends on staff remembering the compatibility rules, which is slow and error-prone in an emergency. Data about donors, camps and hospitals is scattered, which causes duplication and inconsistency.

2.3 PROPOSED SOLUTION

The proposed system stores all information in a single normalized MySQL database. Every donation made at a camp produces a blood unit that is linked to a blood group and a hospital, and the hospital inventory is maintained through triggers rather than manual edits. A view lists, for every request, the eligible donors whose blood group is compatible with the patient's. Match records are generated from this view, and a stored procedure returns the most suitable donor for any request.

2.4 FUNCTIONAL REQUIREMENTS

The system should: register hospitals, donors, patients and donation camps; record eligibility checks and donations; track each blood unit with its collection date, expiry date and status (Available, Reserved or Expired); maintain total, available and reserved units per hospital and blood group; record blood requests with urgency; list compatible donors and match scores for each request; record transfers of units between hospitals; and support queries for eligible donors, emergency requests, expired blood, low inventory and camp-wise donations.

2.5 NON-FUNCTIONAL REQUIREMENTS

The database must preserve integrity through primary keys, foreign keys, NOT NULL and UNIQUE constraints, so that no unit, request or match refers to a record that does not exist. Inventory counts

must stay consistent with unit status changes, and the system should respond quickly to queries on a small to medium sized blood bank network.

2.6 TOOLS USED

MySQL 8 is used for the database, including tables, a view, triggers and a stored procedure, and the MySQL command-line client is used to run and verify the statements.

3. ER MODEL

3.1 ER DIAGRAM

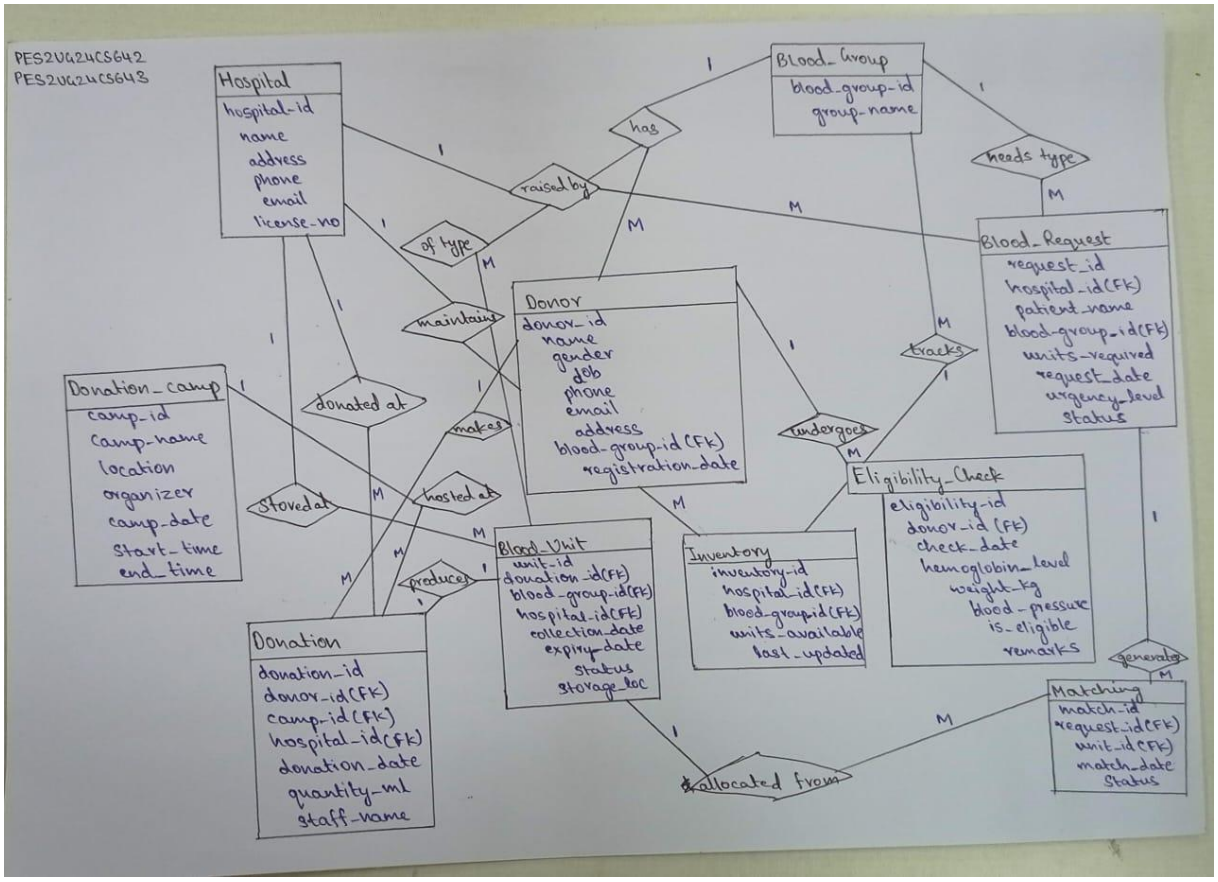


Fig 3.1 Entity-Relationship diagram of the Blood Bank Management and Donor-Recipient Matching System

3.2 ENTITIES

The model contains ten entities. Hospital stores the hospital id, name, address, phone, email and licence number. Blood_Group stores the blood group id and group name. Donor stores personal details, the blood group and the registration date. Eligibility_Check records the haemoglobin level, weight, blood pressure, eligibility result and remarks of each screening. Donation_Camp stores the camp name, location, organizer, date and timings. Donation records each donation with the donor, camp, hospital, date, quantity in millilitres and the staff name. Blood_Unit represents a unit of blood produced from a donation, with its blood group, hospital, collection and expiry dates, status and storage location. Inventory holds the units available for each hospital and blood group. Blood_Request records a hospital's request for a blood group, with units required, request date, urgency level and status. Matching links a request to a blood unit allocated for it.

3.3 RELATIONSHIPS

A hospital raises many blood requests, receives many donations, maintains many inventory records and stores many blood units, so each of these is a one-to-many relationship from Hospital. A blood group is held by many donors, is needed by many requests, is the type of many blood units and is tracked by many inventory records. A donor makes many donations and undergoes many eligibility checks. A donation camp hosts many donations, and each donation produces blood units. Finally, a blood request generates many matches, and a blood unit can be allocated through many matches over time.

4. ER TO RELATIONAL MAPPING

4.1 STEPS OF ALGORITHM FOR CHOOSEN PROBLEM

Step 1 - Regular entity types. Each strong entity (Hospital, Blood_Group, Donor, Donation_Camp, Donation, Eligibility_Check, Blood_Unit, Inventory, Blood_Request and Matching) is mapped to a relation whose attributes are its simple attributes. The identifier of each entity (for example donor_id) becomes the primary key of the relation.

Step 2 - Weak entity types. The model has no weak entity types, so no relation is created in this step.

Step 3 - Binary one-to-one relationships. The model has no one-to-one relationships to map.

Step 4 - Binary one-to-many relationships. For each one-to-many relationship, the primary key of the relation on the one side is added as a foreign key to the relation on the many side. This gives hospital_id in Blood_Request, Donation, Blood_Unit and Inventory; blood_group_id in Donor, Blood_Request, Blood_Unit and Inventory; donor_id in Donation and Eligibility_Check; camp_id in Donation; donation_id in Blood_Unit; and request_id and unit_id in Matching.

Step 5 - Binary many-to-many relationships. The relationship between blood requests and blood units is resolved through the Matching relation, which holds the two foreign keys together with match_date and status.

Step 6 - Multivalued attributes. The model has no multivalued attributes.

Step 7 - N-ary relationships. The model has no relationship of degree greater than two.

4.2 COMPLETE DIAGRAM OF RELATIONAL MAPPING

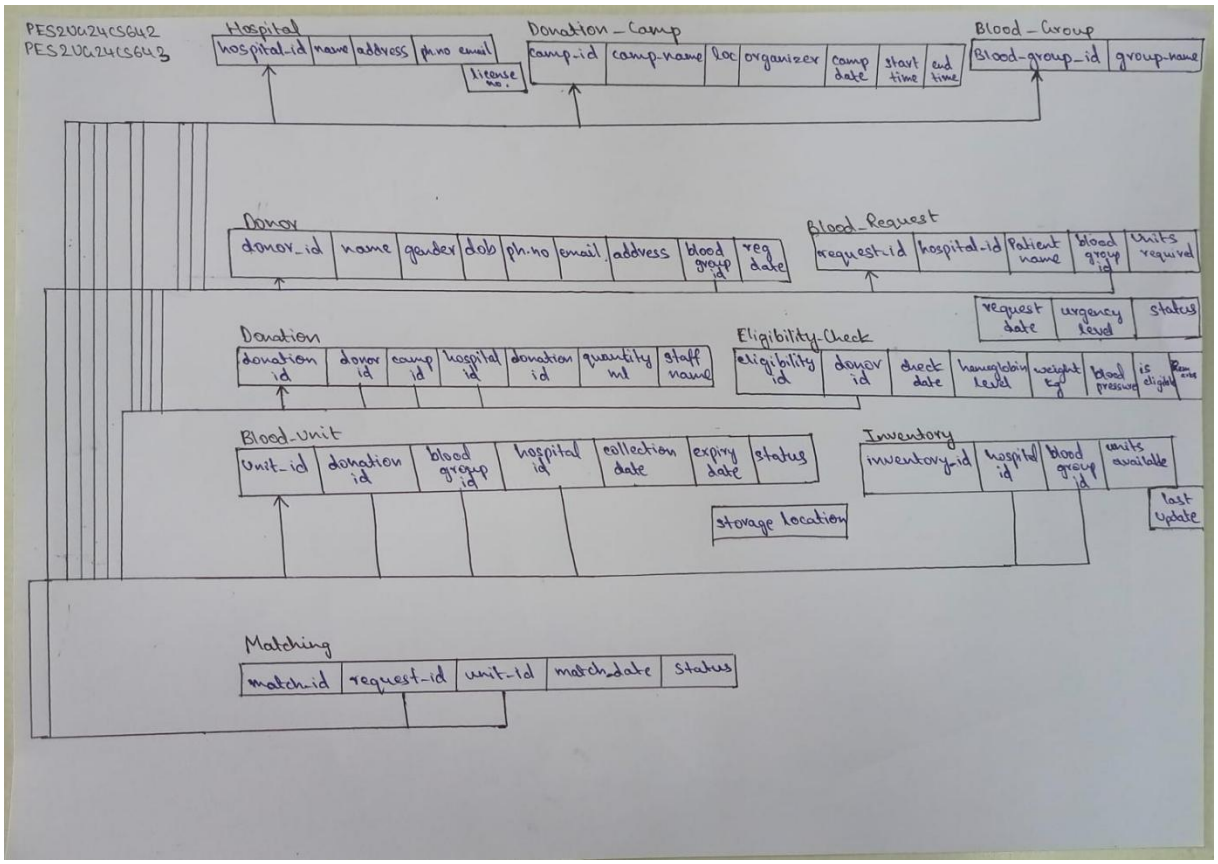


Fig 4.1 Relational schema obtained from the ER diagram

In the implemented MySQL database, the schema was extended with a Patient relation (referenced by Blood_Request) and a Transfer relation for moving units between hospitals. Matching also records the donor, compatibility and match score for each request.

5. DDL STATEMENTS

STATEMENTS WITH SCREEN SHOTS OF THE TABLE CREATION

Database creation

```
mysql> CREATE DATABASE blood_management;
Query OK, 1 row affected (0.06 sec)

mysql> USE blood_management;
Database changed
```

Table creation: Donor, Hospital and Eligibility

```
mysql> CREATE TABLE Donor (
    ->     Donor_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Name VARCHAR(100) NOT NULL,
    ->     DOB DATE NOT NULL,
    ->     Gender VARCHAR(10),
    ->     Blood_Group VARCHAR(5) NOT NULL,
    ->     Phone VARCHAR(15),
    ->     Address VARCHAR(255),
    ->     Last_Donation_Date DATE
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE Hospital (
    ->     Hospital_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Hospital_Name VARCHAR(150) NOT NULL,
    ->     Location VARCHAR(150),
    ->     Contact VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Eligibility (
    ->     Eligibility_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Donor_ID INT NOT NULL,
    ->     Eligible BOOLEAN NOT NULL,
    ->     Reason VARCHAR(255),
    ->     Review_Date DATE,
    ->     FOREIGN KEY (Donor_ID) REFERENCES Donor(Donor_ID)
    -> );
```

Table creation: Donation, Patient and Blood_Request

```
mysql> CREATE TABLE Donation (
    ->     Donation_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Donor_ID INT NOT NULL,
    ->     Donation_Date DATE NOT NULL,
    ->     Units INT NOT NULL,
    ->     Screening_Status VARCHAR(30),
    ->     FOREIGN KEY (Donor_ID) REFERENCES Donor(Donor_ID)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE Patient (
    ->     Patient_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Hospital_ID INT NOT NULL,
    ->     Name VARCHAR(100) NOT NULL,
    ->     Blood_Group VARCHAR(5) NOT NULL,
    ->     Medical_Status VARCHAR(50),
    ->     FOREIGN KEY (Hospital_ID) REFERENCES Hospital(Hospital_ID)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Blood_Request (
    ->     Request_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Patient_ID INT NOT NULL,
    ->     Hospital_ID INT NOT NULL,
    ->     Blood_Group VARCHAR(5) NOT NULL,
    ->     Units_Required INT NOT NULL,
    ->     Urgency VARCHAR(20),
    ->     Request_Date DATE NOT NULL,
    ->     Status VARCHAR(30) DEFAULT 'Pending',
    ->     FOREIGN KEY (Patient_ID) REFERENCES Patient(Patient_ID),
    ->     FOREIGN KEY (Hospital_ID) REFERENCES Hospital(Hospital_ID)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql>
```

Table creation: Blood_Unit, Match_Result and Transfer

```
mysql> CREATE TABLE Blood_Unit (
    ->     Unit_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Donation_ID INT NOT NULL,
    ->     Blood_Group VARCHAR(5) NOT NULL,
    ->     Collection_Date DATE NOT NULL,
    ->     Expiry_Date DATE NOT NULL,
    ->     Status VARCHAR(30) DEFAULT 'Available',
    ->     Hospital_ID INT,
    ->     FOREIGN KEY (Donation_ID) REFERENCES Donation(Donation_ID),
    ->     FOREIGN KEY (Hospital_ID) REFERENCES Hospital(Hospital_ID)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE TABLE Match_Result (
    ->     Match_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Request_ID INT NOT NULL,
    ->     Donor_ID INT NOT NULL,
    ->     Compatibility VARCHAR(30),
    ->     Match_Score DECIMAL(5,2),
    ->     Match_Status VARCHAR(30) DEFAULT 'Suggested',
    ->     FOREIGN KEY (Request_ID) REFERENCES Blood_Request(Request_ID),
    ->     FOREIGN KEY (Donor_ID) REFERENCES Donor(Donor_ID)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE TABLE Transfer (
    ->     Transfer_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     From_Hospital INT NOT NULL,
    ->     To_Hospital INT NOT NULL,
    ->     Unit_ID INT NOT NULL,
    ->     Transfer_Date DATE NOT NULL,
    ->     Status VARCHAR(30) DEFAULT 'Requested',
    ->     FOREIGN KEY (From_Hospital) REFERENCES Hospital(Hospital_ID),
    ->     FOREIGN KEY (To_Hospital) REFERENCES Hospital(Hospital_ID),
    ->     FOREIGN KEY (Unit_ID) REFERENCES Blood_Unit(Unit_ID)
    -> );
Query OK, 0 rows affected (0.10 sec)
```

Tables in the database

```
mysql> SHOW TABLES;
+----------------------------+
| Tables_in_blood_management |
+----------------------------+
| blood_request              |
| blood_unit                 |
| donation                   |
| donor                      |
| eligibility                |
| hospital                   |
| match_result               |
| patient                    |
| transfer                   |
+----------------------------+
9 rows in set (0.03 sec)

mysql>
```

Inserting hospitals

```
mysql> INSERT INTO Hospital (Hospital_Name, Location, Contact) VALUES
    -> ('Apollo Hospital', 'Chennai', '04440010001'),
    -> ('Fortis Hospital', 'Bangalore', '08040010002'),
    -> ('Manipal Hospital', 'Bangalore', '08040010003'),
    -> ('Kauvery Hospital', 'Chennai', '04440010004'),
    -> ('Narayana Health', 'Bangalore', '08040010005');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Hospital;
+-------------+-----------------+-----------+-------------+
| Hospital_ID | Hospital_Name   | Location  | Contact     |
+-------------+-----------------+-----------+-------------+
|           1 | Apollo Hospital | Chennai   | 04440010001 |
|           2 | Fortis Hospital | Bangalore | 08040010002 |
|           3 | Manipal Hospital| Bangalore | 08040010003 |
|           4 | Kauvery Hospital| Chennai   | 04440010004 |
|           5 | Narayana Health | Bangalore | 08040010005 |
+-------------+-----------------+-----------+-------------+
5 rows in set (0.00 sec)
```

Inserting donors

```
mysql> INSERT INTO Donor
    -> (Name, DOB, Gender, Blood_Group, Phone, Address, Last_Donation_Date)
    -> VALUES
    -> ('Rahul Sharma','2001-05-12','Male','O+','9876543210','Chennai','2026
-05-20'),
    -> ('Ananya Rao','2002-08-21','Female','A+','9876543211','Bangalore','20
26-06-15'),
    -> ('Arjun Kumar','2000-03-18','Male','B+','9876543212','Chennai','2026-
04-10'),
    -> ('Sneha Patel','2003-11-05','Female','AB+','9876543213','Bangalore','
2026-08-10'),
    -> ('Vikram Singh','1999-07-22','Male','O-','9876543214','Chennai','2026
-05-01'),
    -> ('Priya Nair','2001-02-14','Female','A-','9876543215','Bangalore','20
26-06-01'),
    -> ('Aditya Mehta','1998-09-30','Male','B-','9876543216','Chennai','2026
-07-01'),
    -> ('Kavya Iyer','2002-12-11','Female','AB-','9876543217','Bangalore',NU
LL),
    -> ('Rohan Das','2000-01-25','Male','O+','9876543218','Chennai','2026-03
-15'),
    -> ('Meera Kapoor','2003-06-18','Female','A+','9876543219','Bangalore',N
ULL);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Donor table contents

```
mysql> SELECT * FROM Donor;
+----------+--------------+------------+--------+-------------+------------+-----------+--------------------+
| Donor_ID | Name         | DOB        | Gender | Blood_Group | Phone      | Address   | Last_Donation_Date |
+----------+--------------+------------+--------+-------------+------------+-----------+--------------------+
|        1 | Rahul Sharma | 2001-05-12 | Male   | O+          | 9876543210 | Chennai   | 2026-05-20         |
|        2 | Ananya Rao   | 2002-08-21 | Female | A+          | 9876543211 | Bangalore | 2026-06-15         |
|        3 | Arjun Kumar  | 2000-03-18 | Male   | B+          | 9876543212 | Chennai   | 2026-04-10         |
|        4 | Sneha Patel  | 2003-11-05 | Female | AB+         | 9876543213 | Bangalore | 2026-08-10         |
|        5 | Vikram Singh | 1999-07-22 | Male   | O-          | 9876543214 | Chennai   | 2026-05-01         |
|        6 | Priya Nair   | 2001-02-14 | Female | A-          | 9876543215 | Bangalore | 2026-06-01         |
|        7 | Aditya Mehta | 1998-09-30 | Male   | B-          | 9876543216 | Chennai   | 2026-07-01         |
|        8 | Kavya Iyer   | 2002-12-11 | Female | AB-         | 9876543217 | Bangalore | NULL               |
|        9 | Rohan Das    | 2000-01-25 | Male   | O+          | 9876543218 | Chennai   | 2026-03-15         |
|       10 | Meera Kapoor | 2003-06-18 | Female | A+          | 9876543219 | Bangalore | NULL               |
+----------+--------------+------------+--------+-------------+------------+-----------+--------------------+
10 rows in set (0.00 sec)
```

Inserting eligibility records

```
mysql> INSERT INTO Eligibility
    -> (Donor_ID, Eligible, Reason, Review_Date)
    -> VALUES
    -> (1,TRUE,'Eligible for donation','2026-08-24'),
    -> (2,TRUE,'Eligible for donation','2026-08-24'),
    -> (3,FALSE,'Minimum donation interval not completed','2026-08-24'),
    -> (4,TRUE,'Eligible for donation','2026-08-24'),
    -> (5,TRUE,'Eligible for donation','2026-08-24'),
    -> (6,TRUE,'Eligible for donation','2026-08-24'),
    -> (7,FALSE,'Temporary medical deferral','2026-09-15'),
    -> (8,TRUE,'Eligible for donation','2026-08-24'),
    -> (9,FALSE,'Minimum donation interval not completed','2026-09-01'),
    -> (10,TRUE,'Eligible for donation','2026-08-24');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Eligibility table contents

```
mysql> SELECT * FROM Eligibility;
+----------------+----------+----------+---------------------------------------+-------------+
| Eligibility_ID | Donor_ID | Eligible | Reason                                | Review_Date |
+----------------+----------+----------+---------------------------------------+-------------+
|              1 |        1 |        1 | Eligible for donation                 | 2026-08-24  |
|              2 |        2 |        1 | Eligible for donation                 | 2026-08-24  |
|              3 |        3 |        0 | Minimum donation interval not completed | 2026-08-24  |
|              4 |        4 |        1 | Eligible for donation                 | 2026-08-24  |
|              5 |        5 |        1 | Eligible for donation                 | 2026-08-24  |
|              6 |        6 |        1 | Eligible for donation                 | 2026-08-24  |
|              7 |        7 |        0 | Temporary medical deferral            | 2026-09-15  |
|              8 |        8 |        1 | Eligible for donation                 | 2026-08-24  |
|              9 |        9 |        0 | Minimum donation interval not completed | 2026-09-01  |
|             10 |       10 |        1 | Eligible for donation                 | 2026-08-24  |
+----------------+----------+----------+---------------------------------------+-------------+
10 rows in set (0.00 sec)
```

Inserting donations and patients

```
mysql> INSERT INTO Donation
    -> (Donor_ID, Donation_Date, Units, Screening_Status)
    -> VALUES
    -> (1,'2026-05-20',1,'Passed'),
    -> (2,'2026-06-15',1,'Passed'),
    -> (3,'2026-04-10',1,'Passed'),
    -> (4,'2026-08-10',1,'Passed'),
    -> (5,'2026-05-01',1,'Passed'),
    -> (6,'2026-06-01',1,'Passed'),
    -> (7,'2026-07-01',1,'Passed'),
    -> (8,'2026-08-12',1,'Passed'),
    -> (9,'2026-03-15',1,'Passed'),
    -> (10,'2026-08-15',1,'Passed');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Patient
    -> (Hospital_ID, Name, Blood_Group, Medical_Status)
    -> VALUES
    -> (1,'Karthik Rao','O+','Critical'),
    -> (2,'Meghana Shah','A+','Stable'),
    -> (3,'Arjun Reddy','B+','Critical'),
    -> (4,'Neha Gupta','AB+','Stable'),
    -> (5,'Ravi Kumar','O-','Critical'),
    -> (1,'Sanjay Patel','A-','Critical');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Patient;
+------------+-------------+--------------+-------------+-----------------+
| Patient_ID | Hospital_ID | Name         | Blood_Group | Medical_Status  |
+------------+-------------+--------------+-------------+-----------------+
|          1 |           1 | Karthik Rao  | O+          | Critical        |
|          2 |           2 | Meghana Shah | A+          | Stable          |
|          3 |           3 | Arjun Reddy  | B+          | Critical        |
|          4 |           4 | Neha Gupta   | AB+         | Stable          |
|          5 |           5 | Ravi Kumar   | O-          | Critical        |
|          6 |           1 | Sanjay Patel | A-          | Critical        |
+------------+-------------+--------------+-------------+-----------------+
6 rows in set (0.00 sec)
```

Inserting blood requests

```
mysql> INSERT INTO Blood_Request
    -> (Patient_ID, Hospital_ID, Blood_Group, Units_Required, Urgency, Request_Date)
    -> VALUES
    -> (1,1,'O+',2,'Emergency','2026-08-24'),
    -> (2,2,'A+',1,'Normal','2026-08-24'),
    -> (3,3,'B+',2,'Emergency','2026-08-24'),
    -> (4,4,'AB+',1,'Normal','2026-08-24'),
    -> (5,5,'O-',3,'Emergency','2026-08-24'),
    -> (6,1,'A-',1,'Urgent','2026-08-24');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Blood_Request;
+------------+------------+-------------+-------------+----------------+-----------+--------------+---------+
| Request_ID | Patient_ID | Hospital_ID | Blood_Group | Units_Required | Urgency   | Request_Date | Status  |
+------------+------------+-------------+-------------+----------------+-----------+--------------+---------+
|          1 |          1 |           1 | O+          |              2 | Emergency | 2026-08-24   | Pending |
|          2 |          2 |           2 | A+          |              1 | Normal    | 2026-08-24   | Pending |
|          3 |          3 |           3 | B+          |              2 | Emergency | 2026-08-24   | Pending |
|          4 |          4 |           4 | AB+         |              1 | Normal    | 2026-08-24   | Pending |
|          5 |          5 |           5 | O-          |              3 | Emergency | 2026-08-24   | Pending |
|          6 |          6 |           1 | A-          |              1 | Urgent    | 2026-08-24   | Pending |
+------------+------------+-------------+-------------+----------------+-----------+--------------+---------+
6 rows in set (0.00 sec)

mysql>
```

Inserting blood units

```
mysql> INSERT INTO Blood_Unit
    -> (Donation_ID, Blood_Group, Collection_Date, Expiry_Date, Status, Hosp
ital_ID)
    -> VALUES
    -> (1,'O+','2026-05-20','2026-07-01','Expired',1),
    -> (2,'A+','2026-06-15','2026-07-27','Expired',2),
    -> (3,'B+','2026-04-10','2026-05-22','Expired',3),
    -> (4,'AB+','2026-08-10','2026-09-21','Available',4),
    -> (5,'O-','2026-05-01','2026-06-12','Expired',5),
    -> (6,'A-','2026-06-01','2026-07-13','Expired',1),
    -> (8,'AB-','2026-08-12','2026-09-23','Available',2),
    -> (10,'A+','2026-08-15','2026-09-26','Available',3);
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

QUERY 1 - SHOW ELIGIBLE DONORS

```
mysql> SELECT
    ->     d.Donor_ID,
    ->     d.Name,
    ->     d.Blood_Group,
    ->     e.Eligible
    -> FROM Donor d
    -> JOIN Eligibility e
    -> ON d.Donor_ID = e.Donor_ID
    -> WHERE e.Eligible = TRUE;
+----------+--------------+-------------+----------+
| Donor_ID | Name         | Blood_Group | Eligible |
+----------+--------------+-------------+----------+
|        1 | Rahul Sharma | O+          |        1 |
|        2 | Ananya Rao   | A+          |        1 |
|        4 | Sneha Patel  | AB+         |        1 |
|        5 | Vikram Singh | O-          |        1 |
|        6 | Priya Nair   | A-          |        1 |
|        8 | Kavya Iyer   | AB-         |        1 |
|       10 | Meera Kapoor | A+          |        1 |
+----------+--------------+-------------+----------+
7 rows in set (0.00 sec)
```

QUERY 2 - FIND EMERGENCY BLOOD REQUESTS

```
mysql> SELECT
    ->     Request_ID,
    ->     Patient_ID,
    ->     Blood_Group,
    ->     Units_Required,
    ->     Urgency
    -> FROM Blood_Request
    -> WHERE Urgency = 'Emergency';
+------------+------------+-------------+---------------+-----------+
| Request_ID | Patient_ID | Blood_Group | Units_Required | Urgency   |
+------------+------------+-------------+---------------+-----------+
|          1 |          1 | O+          |             2 | Emergency |
|          3 |          3 | B+          |             2 | Emergency |
|          5 |          5 | O-          |             3 | Emergency |
+------------+------------+-------------+---------------+-----------+
3 rows in set (0.00 sec)

mysql>
```

QUERY 3 - CURRENT HOSPITAL INVENTORY

```
mysql> SELECT
    ->     h.Hospital_Name,
    ->     b.Blood_Group,
    ->     COUNT(b.Unit_ID) AS Available_Units
    -> FROM Blood_Unit b
    -> JOIN Hospital h
    -> ON b.Hospital_ID = h.Hospital_ID
    -> WHERE b.Status = 'Available'
    -> GROUP BY h.Hospital_Name, b.Blood_Group;
+------------------+-------------+-----------------+
| Hospital_Name    | Blood_Group | Available_Units |
+------------------+-------------+-----------------+
| Kauvery Hospital | AB+         |               1 |
| Fortis Hospital  | AB-         |               1 |
| Manipal Hospital | A+          |               1 |
+------------------+-------------+-----------------+
3 rows in set (0.01 sec)
```

QUERY 4 - FIND EXPIRED BLOOD

```
mysql> SELECT
    ->     Unit_ID,
    ->     Blood_Group,
    ->     Expiry_Date,
    ->     Status
    -> FROM Blood_Unit
    -> WHERE Expiry_Date < CURDATE();
+---------+-------------+-------------+---------+
| Unit_ID | Blood_Group | Expiry_Date | Status  |
+---------+-------------+-------------+---------+
|       1 | O+          | 2026-07-01  | Expired |
|       2 | A+          | 2026-07-27  | Expired |
|       3 | B+          | 2026-05-22  | Expired |
|       5 | O-          | 2026-06-12  | Expired |
|       6 | A-          | 2026-07-13  | Expired |
+---------+-------------+-------------+---------+
5 rows in set (0.00 sec)
```

QUERY 5 - FIND DONORS MATCHING O+

```
mysql> SELECT
    ->     d.Donor_ID,
    ->     d.Name,
    ->     d.Blood_Group
    -> FROM Donor d
    -> JOIN Eligibility e
    -> ON d.Donor_ID = e.Donor_ID
    -> WHERE e.Eligible = TRUE
    -> AND d.Blood_Group IN ('O+','O-');
+----------+--------------+-------------+
| Donor_ID | Name         | Blood_Group |
+----------+--------------+-------------+
|        1 | Rahul Sharma | O+          |
|        5 | Vikram Singh | O-          |
+----------+--------------+-------------+
2 rows in set (0.00 sec)
```

QUERY 6 - FIND DONORS MATCHING A+

```
mysql> SELECT
    ->     d.Donor_ID,
    ->     d.Name,
    ->     d.Blood_Group
    -> FROM Donor d
    -> JOIN Eligibility e
    -> ON d.Donor_ID = e.Donor_ID
    -> WHERE e.Eligible = TRUE
    -> AND d.Blood_Group IN ('A+','A-','O+','O-');
+----------+--------------+-------------+
| Donor_ID | Name         | Blood_Group |
+----------+--------------+-------------+
|        1 | Rahul Sharma | O+          |
|        2 | Ananya Rao   | A+          |
|        5 | Vikram Singh | O-          |
|        6 | Priya Nair   | A-          |
|       10 | Meera Kapoor | A+          |
+----------+--------------+-------------+
5 rows in set (0.00 sec)
```

QUERY 7 - FIND HOSPITALS WITH LOW INVENTORY

```
mysql> SELECT
    ->     h.Hospital_Name,
    ->     b.Blood_Group,
    ->     COUNT(b.Unit_ID) AS Available_Units
    -> FROM Blood_Unit b
    -> JOIN Hospital h
    -> ON b.Hospital_ID = h.Hospital_ID
    -> WHERE b.Status = 'Available'
    -> GROUP BY h.Hospital_Name, b.Blood_Group
    -> HAVING COUNT(b.Unit_ID) < 2;
+-----------------+-------------+-----------------+
| Hospital_Name   | Blood_Group | Available_Units |
+-----------------+-------------+-----------------+
| Kauvery Hospital | AB+         |               1 |
| Fortis Hospital  | AB-         |               1 |
| Manipal Hospital | A+          |               1 |
+-----------------+-------------+-----------------+
3 rows in set (0.00 sec)
```

VIEW COMPATIBLE DONORS

```
mysql> CREATE VIEW Compatible_Donors AS
    -> SELECT
    ->     br.Request_ID,
    ->     br.Blood_Group AS Required_Group,
    ->     d.Donor_ID,
    ->     d.Name,
    ->     d.Blood_Group AS Donor_Group,
    ->     d.Last_Donation_Date
    -> FROM Blood_Request br
    -> JOIN Donor d
    -> JOIN Eligibility e
    -> ON d.Donor_ID = e.Donor_ID
    -> WHERE e.Eligible = TRUE
    -> AND (
    ->     (br.Blood_Group='O+' AND d.Blood_Group IN ('O+','O-')) OR
    ->     (br.Blood_Group='A+' AND d.Blood_Group IN ('A+','A-','O+','O-')) OR
    ->     (br.Blood_Group='B+' AND d.Blood_Group IN ('B+','B-','O+','O-')) OR
    ->     (br.Blood_Group='AB+' AND d.Blood_Group IN ('AB+','AB-','A+','A-','B+','B-','O+','O-')) OR
    ->     (br.Blood_Group='O-' AND d.Blood_Group='O-') OR
    ->     (br.Blood_Group='A-' AND d.Blood_Group IN ('A-','O-')) OR
    ->     (br.Blood_Group='B-' AND d.Blood_Group IN ('B-','O-')) OR
    ->     (br.Blood_Group='AB-' AND d.Blood_Group IN ('AB-','A-','B-','O-'))
    -> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Compatible_Donors;
+------------+----------------+----------+--------------+-------------+--------------------+
| Request_ID | Required_Group | Donor_ID | Name         | Donor_Group | Last_Donation_Date |
+------------+----------------+----------+--------------+-------------+--------------------+
|          1 | O+             |        5 | Vikram Singh | O-          | 2026-05-01         |
|          1 | O+             |        1 | Rahul Sharma | O+          | 2026-05-20         |
|          2 | A+             |       10 | Meera Kapoor | A+          | NULL               |
|          2 | A+             |        6 | Priya Nair   | A-          | 2026-06-01         |
|          2 | A+             |        5 | Vikram Singh | O-          | 2026-05-01         |
|          2 | A+             |        2 | Ananya Rao   | A+          | 2026-06-15         |
|          2 | A+             |        1 | Rahul Sharma | O+          | 2026-05-20         |
|          3 | B+             |        5 | Vikram Singh | O-          | 2026-05-01         |
|          3 | B+             |        1 | Rahul Sharma | O+          | 2026-05-20         |
|          4 | AB+            |       10 | Meera Kapoor | A+          | NULL               |
|          4 | AB+            |        8 | Kavya Iyer   | AB-         | NULL               |
|          4 | AB+            |        6 | Priya Nair   | A-          | 2026-06-01         |
|          4 | AB+            |        5 | Vikram Singh | O-          | 2026-05-01         |
|          4 | AB+            |        4 | Sneha Patel  | AB+         | 2026-08-10         |
|          4 | AB+            |        2 | Ananya Rao   | A+          | 2026-06-15         |
|          4 | AB+            |        1 | Rahul Sharma | O+          | 2026-05-20         |
|          5 | O-             |        5 | Vikram Singh | O-          | 2026-05-01         |
|          6 | A-             |        6 | Priya Nair   | A-          | 2026-06-01         |
|          6 | A-             |        5 | Vikram Singh | O-          | 2026-05-01         |
+------------+----------------+----------+--------------+-------------+--------------------+
19 rows in set (0.01 sec)

mysql>
```

CREATE MATCHING RECORDS

```
mysql> INSERT INTO Match_Result
    -> (Request_ID, Donor_ID, Compatibility, Match_Score, Match_Status)
    ->
    -> SELECT
    ->     Request_ID,
    ->     Donor_ID,
    ->     'Compatible',
    ->     95,
    ->     'Suggested'
    -> FROM Compatible_Donors;
Query OK, 19 rows affected (0.01 sec)
Records: 19  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Match_Result;
+----------+------------+----------+---------------+-------------+--------------+
| Match_ID | Request_ID | Donor_ID | Compatibility | Match_Score | Match_Status |
+----------+------------+----------+---------------+-------------+--------------+
|        1 |          1 |        5 | Compatible    |       95.00 | Suggested    |
|        2 |          1 |        1 | Compatible    |       95.00 | Suggested    |
|        3 |          2 |       10 | Compatible    |       95.00 | Suggested    |
|        4 |          2 |        6 | Compatible    |       95.00 | Suggested    |
|        5 |          2 |        5 | Compatible    |       95.00 | Suggested    |
|        6 |          2 |        2 | Compatible    |       95.00 | Suggested    |
|        7 |          2 |        1 | Compatible    |       95.00 | Suggested    |
|        8 |          3 |        5 | Compatible    |       95.00 | Suggested    |
|        9 |          3 |        1 | Compatible    |       95.00 | Suggested    |
|       10 |          4 |       10 | Compatible    |       95.00 | Suggested    |
|       11 |          4 |        8 | Compatible    |       95.00 | Suggested    |
|       12 |          4 |        6 | Compatible    |       95.00 | Suggested    |
|       13 |          4 |        5 | Compatible    |       95.00 | Suggested    |
|       14 |          4 |        4 | Compatible    |       95.00 | Suggested    |
|       15 |          4 |        2 | Compatible    |       95.00 | Suggested    |
|       16 |          4 |        1 | Compatible    |       95.00 | Suggested    |
|       17 |          5 |        5 | Compatible    |       95.00 | Suggested    |
|       18 |          6 |        6 | Compatible    |       95.00 | Suggested    |
|       19 |          6 |        5 | Compatible    |       95.00 | Suggested    |
+----------+------------+----------+---------------+-------------+--------------+
19 rows in set (0.00 sec)
```

FINDING THE BEST DONOR

```
mysql> CREATE PROCEDURE FindBestDonor(IN req INT)
    -> BEGIN
    ->     SELECT
    ->         d.Donor_ID,
    ->         d.Name,
    ->         d.Blood_Group,
    ->         d.Last_Donation_Date
    ->     FROM Compatible_Donors cd
    ->     JOIN Donor d
    ->     ON cd.Donor_ID=d.Donor_ID
    ->     WHERE cd.Request_ID=req
    ->     ORDER BY d.Last_Donation_Date ASC
    ->     LIMIT 1;
    -> END //
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL FindBestDonor(1);
+----------+--------------+-------------+--------------------+
| Donor_ID | Name         | Blood_Group | Last_Donation_Date |
+----------+--------------+-------------+--------------------+
|        5 | Vikram Singh | O-          | 2026-05-01         |
+----------+--------------+-------------+--------------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql>
```

UPDATE INVENTORY AUTOMATICALLY

```
mysql> CREATE TABLE Inventory (
    ->     Inventory_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Hospital_ID INT NOT NULL,
    ->     Blood_Group VARCHAR(5) NOT NULL,
    ->     Total_Units INT DEFAULT 0,
    ->     Available_Units INT DEFAULT 0,
    ->     Reserved_Units INT DEFAULT 0,
    ->     Last_Updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    ->         ON UPDATE CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (Hospital_ID) REFERENCES Hospital(Hospital_ID),
    ->     UNIQUE (Hospital_ID, Blood_Group)
    -> );
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> INSERT INTO Inventory
    -> (Hospital_ID, Blood_Group, Total_Units, Available_Units, Reserved_Uni
ts)
    -> SELECT
    ->     Hospital_ID,
    ->     Blood_Group,
    ->     COUNT(*) AS Total_Units,
    ->     SUM(CASE WHEN Status = 'Available' THEN 1 ELSE 0 END),
    ->     SUM(CASE WHEN Status = 'Reserved' THEN 1 ELSE 0 END)
    -> FROM Blood_Unit
    -> GROUP BY Hospital_ID, Blood_Group;
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Inventory;
+--------------+-------------+-------------+-------------+-----------------+---------------+---------------------+
| Inventory_ID | Hospital_ID | Blood_Group | Total_Units | Available_Units | Reserved_Units | Last_Updated        |
+--------------+-------------+-------------+-------------+-----------------+---------------+---------------------+
|            1 |           1 | O+          |           1 |               0 |             0 | 2026-08-24 14:18:03 |
|            2 |           2 | A+          |           1 |               0 |             0 | 2026-08-24 14:18:03 |
|            3 |           3 | B+          |           1 |               0 |             0 | 2026-08-24 14:18:03 |
|            4 |           4 | AB+         |           1 |               1 |             0 | 2026-08-24 14:18:03 |
|            5 |           5 | O-          |           1 |               0 |             0 | 2026-08-24 14:18:03 |
|            6 |           1 | A-          |           1 |               0 |             0 | 2026-08-24 14:18:03 |
|            7 |           2 | AB-         |           1 |               1 |             0 | 2026-08-24 14:18:03 |
|            8 |           3 | A+          |           1 |               1 |             0 | 2026-08-24 14:18:03 |
+--------------+-------------+-------------+-------------+-----------------+---------------+---------------------+
8 rows in set (0.00 sec)

mysql>
```

```
mysql> SHOW TABLES;
+----------------------------+
| Tables_in_blood_management |
+----------------------------+
| blood_request              |
| blood_unit                 |
| compatible_donors          |
| donation                   |
| donor                      |
| eligibility                |
| hospital                   |
| inventory                  |
| match_result               |
| patient                    |
| transfer                   |
+----------------------------+
11 rows in set (0.00 sec)
```

```
mysql> DROP TRIGGER IF EXISTS UpdateInventory;
Query OK, 0 rows affected, 1 warning (0.01 sec)
```

```
mysql>
mysql> CREATE TRIGGER UpdateInventory
    -> AFTER UPDATE ON Blood_Unit
    -> FOR EACH ROW
    -> BEGIN
    ->
    ->     IF OLD.Status <> NEW.Status THEN
    ->
    ->         IF OLD.Status = 'Available' THEN
    ->             UPDATE Inventory
    ->             SET Available_Units = Available_Units - 1
    ->             WHERE Hospital_ID = OLD.Hospital_ID
    ->               AND Blood_Group = OLD.Blood_Group;
    ->         END IF;
    ->
    ->         IF OLD.Status = 'Reserved' THEN
    ->             UPDATE Inventory
    ->             SET Reserved_Units = Reserved_Units - 1
    ->             WHERE Hospital_ID = OLD.Hospital_ID
    ->               AND Blood_Group = OLD.Blood_Group;
    ->         END IF;
    ->
    ->         IF NEW.Status = 'Available' THEN
    ->             UPDATE Inventory
    ->             SET Available_Units = Available_Units + 1
    ->             WHERE Hospital_ID = NEW.Hospital_ID
    ->               AND Blood_Group = NEW.Blood_Group;
    ->         END IF;
    ->
    ->         IF NEW.Status = 'Reserved' THEN
    ->             UPDATE Inventory
    ->             SET Reserved_Units = Reserved_Units + 1
    ->             WHERE Hospital_ID = NEW.Hospital_ID
    ->               AND Blood_Group = NEW.Blood_Group;
    ->         END IF;
    ->
    ->     END IF;
    ->
    -> END //
Query OK, 0 rows affected (0.01 sec)
```

SHOWING TRIGGERS



TESTING THE AUTOMATIC INVENTORY UPDATE

```
mysql> SELECT * FROM Inventory
    -> WHERE Hospital_ID = 3
    -> AND Blood_Group = 'A+';
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
| Inventory_ID | Hospital_ID | Blood_Group | Total_Units | Available_Units | Reserved_Units | Last_Updated        |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
|            8 |           3 | A+          |           1 |               1 |              0 | 2026-08-24 14:18:03 |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
1 row in set (0.00 sec)
```

```
mysql> UPDATE Blood_Unit
    -> SET Status = 'Reserved'
    -> WHERE Unit_ID = 8;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM Inventory
    -> WHERE Hospital_ID = 3
    -> AND Blood_Group = 'A+';
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
| Inventory_ID | Hospital_ID | Blood_Group | Total_Units | Available_Units | Reserved_Units | Last_Updated        |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
|            8 |           3 | A+          |           1 |               0 |              1 | 2026-08-24 14:23:03 |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
1 row in set (0.00 sec)
```

The number of available units changed from 1 to 0 - now testing the reverse

```
mysql> UPDATE Blood_Unit
    -> SET Status = 'Available'
    -> WHERE Unit_ID = 8;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM Inventory
    -> WHERE Hospital_ID = 3
    -> AND Blood_Group = 'A+';
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
| Inventory_ID | Hospital_ID | Blood_Group | Total_Units | Available_Units | Reserved_Units | Last_Updated        |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
|            8 |           3 | A+          |           1 |               1 |              0 | 2026-08-24 14:25:58 |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
1 row in set (0.00 sec)
```

The number of available units has changed from 0 to 1

```
mysql> CREATE TABLE Donation_Camp (
    ->     Camp_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Camp_Name VARCHAR(150) NOT NULL,
    ->     Location VARCHAR(150),
    ->     Organizer VARCHAR(100),
    ->     Camp_Date DATE NOT NULL,
    ->     Start_Time TIME,
    ->     End_Time TIME,
    ->     Hospital_ID INT NOT NULL,
    ->     FOREIGN KEY (Hospital_ID) REFERENCES Hospital(Hospital_ID)
    -> );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> INSERT INTO Donation_Camp
    -> (Camp_Name, Location, Organizer, Camp_Date, Start_Time, End_Time, Hospital_ID)
    -> VALUES
    -> ('PESU Blood Donation Drive',
    ->  'Bangalore',
    ->  'PES University',
    ->  '2026-09-05',
    ->  '09:00:00',
    ->  '16:00:00',
    ->  2),
    ->
    -> ('Life Saver Blood Camp',
    ->  'Chennai',
    ->  'Apollo Hospital',
    ->  '2026-09-12',
    ->  '10:00:00',
    ->  '17:00:00',
    ->  1),
    ->
    -> ('Community Blood Donation Camp',
    ->  'Bangalore',
    ->  'Narayana Health',
    ->  '2026-09-20',
    ->  '09:00:00',
    ->  '15:00:00',
    ->  5),
    ->
    -> ('Emergency Blood Donation Drive',
    ->  'Chennai',
    ->  'Kauvery Hospital',
    ->  '2026-10-03',
    ->  '09:30:00',
    ->  '16:30:00',
    ->  4);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Donation_Camp;
+---------+-------------------------------+-----------+-----------------+------------+------------+----------+-------------+
| Camp_ID | Camp_Name                     | Location  | Organizer       | Camp_Date  | Start_Time | End_Time | Hospital_ID |
+---------+-------------------------------+-----------+-----------------+------------+------------+----------+-------------+
|       1 | PESU Blood Donation Drive     | Bangalore | PES University  | 2026-09-05 | 09:00:00   | 16:00:00 |           2 |
|       2 | Life Saver Blood Camp         | Chennai   | Apollo Hospital | 2026-09-12 | 10:00:00   | 17:00:00 |           1 |
|       3 | Community Blood Donation Camp | Bangalore | Narayana Health | 2026-09-20 | 09:00:00   | 15:00:00 |           5 |
|       4 | Emergency Blood Donation Drive| Chennai   | Kauvery Hospital| 2026-10-03 | 09:30:00   | 16:30:00 |           4 |
+---------+-------------------------------+-----------+-----------------+------------+------------+----------+-------------+
4 rows in set (0.00 sec)
```

ADDING CAMP ID AND CREATING FOREIGN KEY

```
mysql> ALTER TABLE Donation
    -> ADD Camp_ID INT;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE Donation
    -> ADD CONSTRAINT fk_donation_camp
    -> FOREIGN KEY (Camp_ID)
    -> REFERENCES Donation_Camp(Camp_ID);
Query OK, 10 rows affected (0.14 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

ASSIGN EXISTING DONATIONS TO CAMPS

```
mysql> UPDATE Donation
    -> SET Camp_ID = 1
    -> WHERE Donation_ID IN (1,2,3);
Query OK, 3 rows affected (0.01 sec)
Rows matched: 3  Changed: 3  Warnings: 0

mysql> UPDATE Donation
    -> SET Camp_ID = 2
    -> WHERE Donation_ID IN (4,5);
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> UPDATE Donation
    -> SET Camp_ID = 3
    -> WHERE Donation_ID IN (6,7,8);
Query OK, 3 rows affected (0.01 sec)
Rows matched: 3  Changed: 3  Warnings: 0

mysql> UPDATE Donation
    -> SET Camp_ID = 4
    -> WHERE Donation_ID IN (9,10);
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql>
```

```
mysql> SELECT * FROM Donation;
+-------------+----------+---------------+-------+------------------+---------+
| Donation_ID | Donor_ID | Donation_Date | Units | Screening_Status | Camp_ID |
+-------------+----------+---------------+-------+------------------+---------+
|           1 |        1 | 2026-05-20    |     1 | Passed           |       1 |
|           2 |        2 | 2026-06-15    |     1 | Passed           |       1 |
|           3 |        3 | 2026-04-10    |     1 | Passed           |       1 |
|           4 |        4 | 2026-08-10    |     1 | Passed           |       2 |
|           5 |        5 | 2026-05-01    |     1 | Passed           |       2 |
|           6 |        6 | 2026-06-01    |     1 | Passed           |       3 |
|           7 |        7 | 2026-07-01    |     1 | Passed           |       3 |
|           8 |        8 | 2026-08-12    |     1 | Passed           |       3 |
|           9 |        9 | 2026-03-15    |     1 | Passed           |       4 |
|          10 |       10 | 2026-08-15    |     1 | Passed           |       4 |
+-------------+----------+---------------+-------+------------------+---------+
10 rows in set (0.00 sec)
```

SHOWING WHICH DONOR TO WHICH CAMP

```
mysql> SELECT
    ->     d.Donation_ID,
    ->     dn.Name AS Donor_Name,
    ->     dc.Camp_Name,
    ->     dc.Location,
    ->     d.Donation_Date,
    ->     d.Units
    -> FROM Donation d
    -> JOIN Donor dn
    ->     ON d.Donor_ID = dn.Donor_ID
    -> JOIN Donation_Camp dc
    ->     ON d.Camp_ID = dc.Camp_ID;
+-------------+-------------+-------------------------------+-----------+---------------+-------+
| Donation_ID | Donor_Name  | Camp_Name                     | Location  | Donation_Date | Units |
+-------------+-------------+-------------------------------+-----------+---------------+-------+
|           1 | Rahul Sharma | PESU Blood Donation Drive     | Bangalore | 2026-05-20    |     1 |
|           2 | Ananya Rao  | PESU Blood Donation Drive     | Bangalore | 2026-06-15    |     1 |
|           3 | Arjun Kumar | PESU Blood Donation Drive     | Bangalore | 2026-04-10    |     1 |
|           4 | Sneha Patel | Life Saver Blood Camp         | Chennai   | 2026-08-10    |     1 |
|           5 | Vikram Singh | Life Saver Blood Camp        | Chennai   | 2026-05-01    |     1 |
|           6 | Priya Nair  | Community Blood Donation Camp | Bangalore | 2026-06-01    |     1 |
|           7 | Aditya Mehta | Community Blood Donation Camp | Bangalore | 2026-07-01    |     1 |
|           8 | Kavya Iyer  | Community Blood Donation Camp | Bangalore | 2026-08-12    |     1 |
|           9 | Rohan Das   | Emergency Blood Donation Drive | Chennai  | 2026-03-15    |     1 |
|          10 | Meera Kapoor | Emergency Blood Donation Drive | Chennai | 2026-08-15    |     1 |
+-------------+-------------+-------------------------------+-----------+---------------+-------+
10 rows in set (0.01 sec)
```

```
mysql> SHOW TABLES;
+-----------------------------+
| Tables_in_blood_management |
+-----------------------------+
| blood_request               |
| blood_unit                  |
| compatible_donors           |
| donation                    |
| donation_camp               |
| donor                       |
| eligibility                 |
| hospital                    |
| inventory                   |
| match_result                |
| patient                     |
| transfer                    |
+-----------------------------+
12 rows in set (0.00 sec)
```

CONNECTING DONOR TO BLOOD GROUP

```
mysql> CREATE TABLE Blood_Group (
    ->     Blood_Group_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Blood_Type VARCHAR(5) NOT NULL UNIQUE,
    ->     Description VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO Blood_Group (Blood_Type, Description) VALUES
    -> ('A+', 'A Positive'),
    -> ('A-', 'A Negative'),
    -> ('B+', 'B Positive'),
    -> ('B-', 'B Negative'),
    -> ('AB+', 'AB Positive'),
    -> ('AB-', 'AB Negative'),
    -> ('O+', 'O Positive'),
    -> ('O-', 'O Negative');
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Blood_Group;
+---------------+------------+-------------+
| Blood_Group_ID | Blood_Type | Description |
+---------------+------------+-------------+
|              1 | A+         | A Positive  |
|              2 | A-         | A Negative  |
|              3 | B+         | B Positive  |
|              4 | B-         | B Negative  |
|              5 | AB+        | AB Positive |
|              6 | AB-        | AB Negative |
|              7 | O+         | O Positive  |
|              8 | O-         | O Negative  |
+---------------+------------+-------------+
8 rows in set (0.00 sec)

mysql>
```

```
mysql> ALTER TABLE Donor
    -> ADD Blood_Group_ID INT;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Donor d
    -> JOIN Blood_Group bg
    -> ON d.Blood_Group = bg.Blood_Type
    -> SET d.Blood_Group_ID = bg.Blood_Group_ID;
Query OK, 10 rows affected (0.01 sec)
Rows matched: 10  Changed: 10  Warnings: 0

mysql> ALTER TABLE Donor
    -> ADD CONSTRAINT fk_donor_blood_group
    -> FOREIGN KEY (Blood_Group_ID)
    -> REFERENCES Blood_Group(Blood_Group_ID);
Query OK, 10 rows affected (0.09 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT
    ->     d.Donor_ID,
    ->     d.Name,
    ->     bg.Blood_Type
    -> FROM Donor d
    -> JOIN Blood_Group bg
    -> ON d.Blood_Group_ID = bg.Blood_Group_ID;
+----------+--------------+------------+
| Donor_ID | Name         | Blood_Type |
+----------+--------------+------------+
|        6 | Priya Nair   | A-         |
|        2 | Ananya Rao   | A+         |
|       10 | Meera Kapoor | A+         |
|        8 | Kavya Iyer   | AB-        |
|        4 | Sneha Patel  | AB+        |
|        7 | Aditya Mehta | B-         |
|        3 | Arjun Kumar  | B+         |
|        5 | Vikram Singh | O-         |
|        1 | Rahul Sharma | O+         |
|        9 | Rohan Das    | O+         |
+----------+--------------+------------+
10 rows in set (0.00 sec)

mysql>
```

CONNECTING BLOOD REQUEST

```
mysql> ALTER TABLE Blood_Request
    -> ADD Blood_Group_ID INT;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Blood_Request br
    -> JOIN Blood_Group bg
    -> ON br.Blood_Group = bg.Blood_Type
    -> SET br.Blood_Group_ID = bg.Blood_Group_ID;
Query OK, 6 rows affected (0.01 sec)
Rows matched: 6  Changed: 6  Warnings: 0

mysql> ALTER TABLE Blood_Request
    -> ADD CONSTRAINT fk_request_blood_group
    -> FOREIGN KEY (Blood_Group_ID)
    -> REFERENCES Blood_Group(Blood_Group_ID);
Query OK, 6 rows affected (0.15 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT
    ->     br.Request_ID,
    ->     bg.Blood_Type,
    ->     br.Units_Required,
    ->     br.Urgency
    -> FROM Blood_Request br
    -> JOIN Blood_Group bg
    -> ON br.Blood_Group_ID = bg.Blood_Group_ID;
+------------+------------+----------------+-----------+
| Request_ID | Blood_Type | Units_Required | Urgency   |
+------------+------------+----------------+-----------+
|          1 | O+         |              2 | Emergency |
|          2 | A+         |              1 | Normal    |
|          3 | B+         |              2 | Emergency |
|          4 | AB+        |              1 | Normal    |
|          5 | O-         |              3 | Emergency |
|          6 | A-         |              1 | Urgent    |
+------------+------------+----------------+-----------+
6 rows in set (0.00 sec)

mysql>
```

CONNECT BLOOD UNIT

```
mysql> ALTER TABLE Blood_Unit
    -> ADD Blood_Group_ID INT;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Blood_Unit bu
    -> JOIN Blood_Group bg
    -> ON bu.Blood_Group = bg.Blood_Type
    -> SET bu.Blood_Group_ID = bg.Blood_Group_ID;
Query OK, 8 rows affected (0.01 sec)
Rows matched: 8  Changed: 8  Warnings: 0

mysql> ALTER TABLE Blood_Unit
    -> ADD CONSTRAINT fk_unit_blood_group
    -> FOREIGN KEY (Blood_Group_ID)
    -> REFERENCES Blood_Group(Blood_Group_ID);
Query OK, 8 rows affected (0.14 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> SELECT
    ->     bu.Unit_ID,
    ->     bg.Blood_Type,
    ->     bu.Status
    -> FROM Blood_Unit bu
    -> JOIN Blood_Group bg
    -> ON bu.Blood_Group_ID = bg.Blood_Group_ID;
+---------+------------+-----------+
| Unit_ID | Blood_Type | Status    |
+---------+------------+-----------+
|       1 | O+         | Expired   |
|       2 | A+         | Expired   |
|       3 | B+         | Expired   |
|       4 | AB+        | Available |
|       5 | O-         | Expired   |
|       6 | A-         | Expired   |
|       7 | AB-        | Available |
|       8 | A+         | Available |
+---------+------------+-----------+
8 rows in set (0.00 sec)
```

CONNECT INVENTORY

```
mysql> ALTER TABLE Inventory
    -> ADD Blood_Group_ID INT;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Inventory i
    -> JOIN Blood_Group bg
    -> ON i.Blood_Group = bg.Blood_Type
    -> SET i.Blood_Group_ID = bg.Blood_Group_ID;
Query OK, 8 rows affected (0.01 sec)
Rows matched: 8  Changed: 8  Warnings: 0

mysql> ALTER TABLE Inventory
    -> ADD CONSTRAINT fk_inventory_blood_group
    -> FOREIGN KEY (Blood_Group_ID)
    -> REFERENCES Blood_Group(Blood_Group_ID);
Query OK, 8 rows affected (0.09 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> SELECT
    ->     i.Inventory_ID,
    ->     h.Hospital_Name,
    ->     bg.Blood_Type,
    ->     i.Total_Units,
    ->     i.Available_Units,
    ->     i.Reserved_Units
    -> FROM Inventory i
    -> JOIN Hospital h
    -> ON i.Hospital_ID = h.Hospital_ID
    -> JOIN Blood_Group bg
    -> ON i.Blood_Group_ID = bg.Blood_Group_ID;
+--------------+------------------+------------+-------------+-----------------+----------------+
| Inventory_ID | Hospital_Name    | Blood_Type | Total_Units | Available_Units | Reserved_Units |
+--------------+------------------+------------+-------------+-----------------+----------------+
|            6 | Apollo Hospital  | A-         |           1 |               0 |              0 |
|            1 | Apollo Hospital  | O+         |           1 |               0 |              0 |
|            2 | Fortis Hospital  | A+         |           1 |               0 |              0 |
|            7 | Fortis Hospital  | AB-        |           1 |               1 |              0 |
|            8 | Manipal Hospital | A+         |           1 |               1 |              0 |
|            3 | Manipal Hospital | B+         |           1 |               0 |              0 |
|            4 | Kauvery Hospital | AB+        |           1 |               1 |              0 |
|            5 | Narayana Health  | O-         |           1 |               0 |              0 |
+--------------+------------------+------------+-------------+-----------------+----------------+
8 rows in set (0.00 sec)
```

VERIFYING THE RELATIONSHIPS

```
mysql> SHOW CREATE TABLE Donor;
+-------+-------------------
```

MATCHING ALIGNMENT

```
mysql> RENAME TABLE Match_Result TO Matching;
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
+-----------------------------+
| Tables_in_blood_management |
+-----------------------------+
| blood_group                 |
| blood_request               |
| blood_unit                  |
| compatible_donors           |
| donation                    |
| donation_camp               |
| donor                       |
| eligibility                 |
| hospital                    |
| inventory                   |
| matching                    |
| patient                     |
| transfer                    |
+-----------------------------+
13 rows in set (0.00 sec)

mysql> DESC Matching;
+----------------+--------------+------+-----+-----------+----------------+
| Field          | Type         | Null | Key | Default   | Extra          |
+----------------+--------------+------+-----+-----------+----------------+
| Match_ID       | int          | NO   | PRI | NULL      | auto_increment |
| Request_ID     | int          | NO   | MUL | NULL      |                |
| Donor_ID       | int          | NO   | MUL | NULL      |                |
| Compatibility | varchar(30)  | YES  |     | NULL      |                |
| Match_Score    | decimal(5,2) | YES  |     | NULL      |                |
| Match_Status   | varchar(30)  | YES  |     | Suggested |                |
+----------------+--------------+------+-----+-----------+----------------+
6 rows in set (0.01 sec)
```

```
mysql> ALTER TABLE Matching
    -> ADD Match_Date DATE;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE Matching
    -> RENAME COLUMN Match_Status TO Status;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Matching
    -> SET Match_Date = '2026-08-24';
Query OK, 19 rows affected (0.01 sec)
Rows matched: 19  Changed: 19  Warnings: 0

mysql> SELECT * FROM Matching;
+----------+------------+----------+---------------+-------------+-----------+------------+
| Match_ID | Request_ID | Donor_ID | Compatibility | Match_Score | Status    | Match_Date |
+----------+------------+----------+---------------+-------------+-----------+------------+
|        1 |          1 |        5 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|        2 |          1 |        1 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|        3 |          2 |       10 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|        4 |          2 |        6 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|        5 |          2 |        5 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|        6 |          2 |        2 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|        7 |          2 |        1 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|        8 |          3 |        5 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|        9 |          3 |        1 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       10 |          4 |       10 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       11 |          4 |        8 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       12 |          4 |        6 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       13 |          4 |        5 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       14 |          4 |        4 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       15 |          4 |        2 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       16 |          4 |        1 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       17 |          5 |        5 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       18 |          6 |        6 | Compatible    |       95.00 | Suggested | 2026-08-24 |
|       19 |          6 |        5 | Compatible    |       95.00 | Suggested | 2026-08-24 |
+----------+------------+----------+---------------+-------------+-----------+------------+
19 rows in set (0.00 sec)
```

ADDING FOREIGN KEY RELATIONSHIPS

```
mysql> ALTER TABLE Matching
    -> ADD CONSTRAINT fk_matching_request
    -> FOREIGN KEY (Request_ID)
    -> REFERENCES Blood_Request(Request_ID);
Query OK, 19 rows affected (0.10 sec)
Records: 19  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE Matching
    -> ADD CONSTRAINT fk_matching_donor
    -> FOREIGN KEY (Donor_ID)
    -> REFERENCES Donor(Donor_ID);
Query OK, 19 rows affected (0.09 sec)
Records: 19  Duplicates: 0  Warnings: 0

mysql>
```

MATCH SCORE

```
mysql> SELECT
    ->     m.Match_ID,
    ->     m.Request_ID,
    ->     d.Name AS Donor_Name,
    ->     d.Blood_Group AS Donor_Blood_Group,
    ->     br.Blood_Group AS Required_Blood_Group,
    ->     m.Match_Score,
    ->     m.Match_Date,
    ->     m.Status
    -> FROM Matching m
    -> JOIN Donor d
    ->     ON m.Donor_ID = d.Donor_ID
    -> JOIN Blood_Request br
    ->     ON m.Request_ID = br.Request_ID
    -> ORDER BY m.Match_Score DESC;
+----------+------------+--------------+-------------------+----------------------+-------------+------------+-----------+
| Match_ID | Request_ID | Donor_Name   | Donor_Blood_Group | Required_Blood_Group | Match_Score | Match_Date | Status    |
+----------+------------+--------------+-------------------+----------------------+-------------+------------+-----------+
|        1 |          1 | Vikram Singh | O-                | O+                   |       95.00 | 2026-08-24 | Suggested |
|        2 |          1 | Rahul Sharma | O+                | O+                   |       95.00 | 2026-08-24 | Suggested |
|        3 |          2 | Meera Kapoor | A+                | A+                   |       95.00 | 2026-08-24 | Suggested |
|        4 |          2 | Priya Nair   | A-                | A+                   |       95.00 | 2026-08-24 | Suggested |
|        5 |          2 | Vikram Singh | O-                | A+                   |       95.00 | 2026-08-24 | Suggested |
|        6 |          2 | Ananya Rao   | A+                | A+                   |       95.00 | 2026-08-24 | Suggested |
|        7 |          2 | Rahul Sharma | O+                | A+                   |       95.00 | 2026-08-24 | Suggested |
|        8 |          3 | Vikram Singh | O-                | B+                   |       95.00 | 2026-08-24 | Suggested |
|        9 |          3 | Rahul Sharma | O+                | B+                   |       95.00 | 2026-08-24 | Suggested |
|       10 |          4 | Meera Kapoor | A+                | AB+                  |       95.00 | 2026-08-24 | Suggested |
|       11 |          4 | Kavya Iyer   | AB-               | AB+                  |       95.00 | 2026-08-24 | Suggested |
|       12 |          4 | Priya Nair   | A-                | AB+                  |       95.00 | 2026-08-24 | Suggested |
|       13 |          4 | Vikram Singh | O-                | AB+                  |       95.00 | 2026-08-24 | Suggested |
|       14 |          4 | Sneha Patel  | AB+               | AB+                  |       95.00 | 2026-08-24 | Suggested |
|       15 |          4 | Ananya Rao   | A+                | AB+                  |       95.00 | 2026-08-24 | Suggested |
|       16 |          4 | Rahul Sharma | O+                | AB+                  |       95.00 | 2026-08-24 | Suggested |
|       17 |          5 | Vikram Singh | O-                | O-                   |       95.00 | 2026-08-24 | Suggested |
|       18 |          6 | Priya Nair   | A-                | A-                   |       95.00 | 2026-08-24 | Suggested |
|       19 |          6 | Vikram Singh | O-                | A-                   |       95.00 | 2026-08-24 | Suggested |
+----------+------------+--------------+-------------------+----------------------+-------------+------------+-----------+
19 rows in set (0.00 sec)

mysql>
```

FINDING BEST DONOR FOR EACH REQUEST

```
mysql> SELECT
    ->     m.Request_ID,
    ->     d.Name AS Donor_Name,
    ->     d.Blood_Group,
    ->     m.Match_Score,
    ->     m.Status
    -> FROM Matching m
    -> JOIN Donor d
    ->     ON m.Donor_ID = d.Donor_ID
    -> WHERE m.Match_Score = (
    ->     SELECT MAX(m2.Match_Score)
    ->     FROM Matching m2
    ->     WHERE m2.Request_ID = m.Request_ID
    -> );
+------------+--------------+-------------+-------------+-----------+
| Request_ID | Donor_Name   | Blood_Group | Match_Score | Status    |
+------------+--------------+-------------+-------------+-----------+
|          1 | Vikram Singh | O-          |       95.00 | Suggested |
|          1 | Rahul Sharma | O+          |       95.00 | Suggested |
|          2 | Meera Kapoor | A+          |       95.00 | Suggested |
|          2 | Priya Nair   | A-          |       95.00 | Suggested |
|          2 | Vikram Singh | O-          |       95.00 | Suggested |
|          2 | Ananya Rao   | A+          |       95.00 | Suggested |
|          2 | Rahul Sharma | O+          |       95.00 | Suggested |
|          3 | Vikram Singh | O-          |       95.00 | Suggested |
|          3 | Rahul Sharma | O+          |       95.00 | Suggested |
|          4 | Meera Kapoor | A+          |       95.00 | Suggested |
|          4 | Kavya Iyer   | AB-         |       95.00 | Suggested |
|          4 | Priya Nair   | A-          |       95.00 | Suggested |
|          4 | Vikram Singh | O-          |       95.00 | Suggested |
|          4 | Sneha Patel  | AB+         |       95.00 | Suggested |
|          4 | Ananya Rao   | A+          |       95.00 | Suggested |
|          4 | Rahul Sharma | O+          |       95.00 | Suggested |
|          5 | Vikram Singh | O-          |       95.00 | Suggested |
|          6 | Priya Nair   | A-          |       95.00 | Suggested |
|          6 | Vikram Singh | O-          |       95.00 | Suggested |
+------------+--------------+-------------+-------------+-----------+
19 rows in set (0.00 sec)
```

```
mysql> SHOW TABLES;
+-----------------------------+
| Tables_in_blood_management |
+-----------------------------+
| blood_group                 |
| blood_request               |
| blood_unit                  |
| compatible_donors           |
| donation                    |
| donation_camp               |
| donor                       |
| eligibility                 |
| hospital                    |
| inventory                   |
| matching                    |
| patient                     |
| transfer                    |
+-----------------------------+
13 rows in set (0.00 sec)

mysql>
```

FINAL TESTING

```
mysql> SHOW TABLES;
+----------------------------+
| Tables_in_blood_management |
+----------------------------+
| blood_group                |
| blood_request              |
| blood_unit                 |
| compatible_donors          |
| donation                   |
| donation_camp              |
| donor                      |
| eligibility                |
| hospital                   |
| inventory                  |
| matching                   |
| patient                    |
| transfer                   |
+----------------------------+
13 rows in set (0.00 sec)
```

TEST BLOOD GROUPS

```
mysql> SELECT * FROM Blood_Group;
+----------------+------------+-------------+
| Blood_Group_ID | Blood_Type | Description |
+----------------+------------+-------------+
|              1 | A+         | A Positive  |
|              2 | A-         | A Negative  |
|              3 | B+         | B Positive  |
|              4 | B-         | B Negative  |
|              5 | AB+        | AB Positive |
|              6 | AB-        | AB Negative |
|              7 | O+         | O Positive  |
|              8 | O-         | O Negative  |
+----------------+------------+-------------+
8 rows in set (0.00 sec)
```

TEST DONOR+ BLOOD GROUPS

```
mysql> SELECT
    ->     d.Donor_ID,
    ->     d.Name,
    ->     bg.Blood_Type
    -> FROM Donor d
    -> JOIN Blood_Group bg
    -> ON d.Blood_Group_ID = bg.Blood_Group_ID;
+----------+--------------+------------+
| Donor_ID | Name         | Blood_Type |
+----------+--------------+------------+
|        6 | Priya Nair   | A-         |
|        2 | Ananya Rao   | A+         |
|       10 | Meera Kapoor | A+         |
|        8 | Kavya Iyer   | AB-        |
|        4 | Sneha Patel  | AB+        |
|        7 | Aditya Mehta | B-         |
|        3 | Arjun Kumar  | B+         |
|        5 | Vikram Singh | O-         |
|        1 | Rahul Sharma | O+         |
|        9 | Rohan Das    | O+         |
+----------+--------------+------------+
10 rows in set (0.00 sec)
```

TEST ELIGIBILITY

```
mysql> SELECT
    ->     d.Name,
    ->     d.Blood_Group,
    ->     e.Eligible,
    ->     e.Reason
    -> FROM Donor d
    -> JOIN Eligibility e
    -> ON d.Donor_ID = e.Donor_ID
    -> ORDER BY e.Eligible DESC;
+--------------+-------------+----------+----------------------------------------+
| Name         | Blood_Group | Eligible | Reason                                 |
+--------------+-------------+----------+----------------------------------------+
| Rahul Sharma | O+          |        1 | Eligible for donation                  |
| Ananya Rao   | A+          |        1 | Eligible for donation                  |
| Sneha Patel  | AB+         |        1 | Eligible for donation                  |
| Vikram Singh | O-          |        1 | Eligible for donation                  |
| Priya Nair   | A-          |        1 | Eligible for donation                  |
| Kavya Iyer   | AB-         |        1 | Eligible for donation                  |
| Meera Kapoor | A+          |        1 | Eligible for donation                  |
| Arjun Kumar  | B+          |        0 | Minimum donation interval not completed |
| Aditya Mehta | B-          |        0 | Temporary medical deferral             |
| Rohan Das    | O+          |        0 | Minimum donation interval not completed |
+--------------+-------------+----------+----------------------------------------+
10 rows in set (0.00 sec)
```

TEST BLOOD REQUESTS

```
mysql> SELECT
    ->     br.Request_ID,
    ->     p.Name AS Patient_Name,
    ->     h.Hospital_Name,
    ->     bg.Blood_Type,
    ->     br.Units_Required,
    ->     br.Urgency,
    ->     br.Status
    -> FROM Blood_Request br
    -> JOIN Patient p
    ->     ON br.Patient_ID = p.Patient_ID
    -> JOIN Hospital h
    ->     ON br.Hospital_ID = h.Hospital_ID
    -> JOIN Blood_Group bg
    ->     ON br.Blood_Group_ID = bg.Blood_Group_ID;
+------------+--------------+-----------------+------------+----------------+-----------+---------+
| Request_ID | Patient_Name | Hospital_Name   | Blood_Type | Units_Required | Urgency   | Status  |
+------------+--------------+-----------------+------------+----------------+-----------+---------+
|          1 | Karthik Rao  | Apollo Hospital | O+         |              2 | Emergency | Pending |
|          6 | Sanjay Patel | Apollo Hospital | A-         |              1 | Urgent    | Pending |
|          2 | Meghana Shah | Fortis Hospital | A+         |              1 | Normal    | Pending |
|          3 | Arjun Reddy  | Manipal Hospital| B+         |              2 | Emergency | Pending |
|          4 | Neha Gupta   | Kauvery Hospital| AB+        |              1 | Normal    | Pending |
|          5 | Ravi Kumar   | Narayana Health | O-         |              3 | Emergency | Pending |
+------------+--------------+-----------------+------------+----------------+-----------+---------+
6 rows in set (0.00 sec)
```

TEST HOSPITAL INVENTORY

```
mysql> SELECT
    ->     h.Hospital_Name,
    ->     bg.Blood_Type,
    ->     i.Total_Units,
    ->     i.Available_Units,
    ->     i.Reserved_Units
    -> FROM Inventory i
    -> JOIN Hospital h
    ->     ON i.Hospital_ID = h.Hospital_ID
    -> JOIN Blood_Group bg
    ->     ON i.Blood_Group_ID = bg.Blood_Group_ID
    -> ORDER BY h.Hospital_Name;
+-----------------+------------+-------------+-----------------+----------------+
| Hospital_Name   | Blood_Type | Total_Units | Available_Units | Reserved_Units |
+-----------------+------------+-------------+-----------------+----------------+
| Apollo Hospital | A-         |           1 |               0 |              0 |
| Apollo Hospital | O+         |           1 |               0 |              0 |
| Fortis Hospital | A+         |           1 |               0 |              0 |
| Fortis Hospital | AB-        |           1 |               1 |              0 |
| Kauvery Hospital| AB+        |           1 |               1 |              0 |
| Manipal Hospital| A+         |           1 |               1 |              0 |
| Manipal Hospital| B+         |           1 |               0 |              0 |
| Narayana Health | O-         |           1 |               0 |              0 |
+-----------------+------------+-------------+-----------------+----------------+
8 rows in set (0.00 sec)
```

Testing automatic inventory trigger - done in the above pages - test donation camps

```
mysql> SELECT
    ->     dc.Camp_Name,
    ->     dc.Location,
    ->     dc.Camp_Date,
    ->     h.Hospital_Name,
    ->     COUNT(d.Donation_ID) AS Total_Donations
    -> FROM Donation_Camp dc
    -> JOIN Hospital h
    ->     ON dc.Hospital_ID = h.Hospital_ID
    -> LEFT JOIN Donation d
    ->     ON dc.Camp_ID = d.Camp_ID
    -> GROUP BY
    ->     dc.Camp_ID,
    ->     dc.Camp_Name,
    ->     dc.Location,
    ->     dc.Camp_Date,
    ->     h.Hospital_Name;
+-------------------------------+-----------+------------+------------------+-----------------+
| Camp_Name                     | Location  | Camp_Date  | Hospital_Name    | Total_Donations |
+-------------------------------+-----------+------------+------------------+-----------------+
| PESU Blood Donation Drive     | Bangalore | 2026-09-05 | Fortis Hospital  |               3 |
| Life Saver Blood Camp         | Chennai   | 2026-09-12 | Apollo Hospital  |               2 |
| Community Blood Donation Camp | Bangalore | 2026-09-20 | Narayana Health  |               3 |
| Emergency Blood Donation Drive| Chennai   | 2026-10-03 | Kauvery Hospital |               2 |
+-------------------------------+-----------+------------+------------------+-----------------+
4 rows in set (0.01 sec)
```

TEST COMPATIBLE DONORS

```
mysql> SELECT *
    -> FROM Compatible_Donors;
+------------+----------------+----------+--------------+-------------+--------------------+
| Request_ID | Required_Group | Donor_ID | Name         | Donor_Group | Last_Donation_Date |
+------------+----------------+----------+--------------+-------------+--------------------+
|          1 | O+             |        5 | Vikram Singh | O-          | 2026-05-01         |
|          1 | O+             |        1 | Rahul Sharma | O+          | 2026-05-20         |
|          2 | A+             |       10 | Meera Kapoor | A+          | NULL               |
|          2 | A+             |        6 | Priya Nair   | A-          | 2026-06-01         |
|          2 | A+             |        5 | Vikram Singh | O-          | 2026-05-01         |
|          2 | A+             |        2 | Ananya Rao   | A+          | 2026-06-15         |
|          2 | A+             |        1 | Rahul Sharma | O+          | 2026-05-20         |
|          3 | B+             |        5 | Vikram Singh | O-          | 2026-05-01         |
|          3 | B+             |        1 | Rahul Sharma | O+          | 2026-05-20         |
|          4 | AB+            |       10 | Meera Kapoor | A+          | NULL               |
|          4 | AB+            |        8 | Kavya Iyer   | AB-         | NULL               |
|          4 | AB+            |        6 | Priya Nair   | A-          | 2026-06-01         |
|          4 | AB+            |        5 | Vikram Singh | O-          | 2026-05-01         |
|          4 | AB+            |        4 | Sneha Patel  | AB+         | 2026-08-10         |
|          4 | AB+            |        2 | Ananya Rao   | A+          | 2026-06-15         |
|          4 | AB+            |        1 | Rahul Sharma | O+          | 2026-05-20         |
|          5 | O-             |        5 | Vikram Singh | O-          | 2026-05-01         |
|          6 | A-             |        6 | Priya Nair   | A-          | 2026-06-01         |
|          6 | A-             |        5 | Vikram Singh | O-          | 2026-05-01         |
+------------+----------------+----------+--------------+-------------+--------------------+
19 rows in set (0.00 sec)
```

TEST MATCHING

```
mysql> SELECT
    ->     m.Match_ID,
    ->     m.Request_ID,
    ->     d.Name AS Donor_Name,
    ->     d.Blood_Group AS Donor_Group,
    ->     br.Blood_Group AS Required_Group,
    ->     m.Match_Score,
    ->     m.Match_Date,
    ->     m.Status
    -> FROM Matching m
    -> JOIN Donor d
    ->     ON m.Donor_ID = d.Donor_ID
    -> JOIN Blood_Request br
    ->     ON m.Request_ID = br.Request_ID
    -> ORDER BY m.Match_Score DESC;
+----------+------------+--------------+-------------+----------------+-------------+------------+-----------+
| Match_ID | Request_ID | Donor_Name   | Donor_Group | Required_Group | Match_Score | Match_Date | Status    |
+----------+------------+--------------+-------------+----------------+-------------+------------+-----------+
|        1 |          1 | Vikram Singh | O-          | O+             |       95.00 | 2026-08-24 | Suggested |
|        2 |          1 | Rahul Sharma | O+          | O+             |       95.00 | 2026-08-24 | Suggested |
|        3 |          2 | Meera Kapoor | A+          | A+             |       95.00 | 2026-08-24 | Suggested |
|        4 |          2 | Priya Nair   | A-          | A+             |       95.00 | 2026-08-24 | Suggested |
|        5 |          2 | Vikram Singh | O-          | A+             |       95.00 | 2026-08-24 | Suggested |
|        6 |          2 | Ananya Rao   | A+          | A+             |       95.00 | 2026-08-24 | Suggested |
|        7 |          2 | Rahul Sharma | O+          | A+             |       95.00 | 2026-08-24 | Suggested |
|        8 |          3 | Vikram Singh | O-          | B+             |       95.00 | 2026-08-24 | Suggested |
|        9 |          3 | Rahul Sharma | O+          | B+             |       95.00 | 2026-08-24 | Suggested |
|       10 |          4 | Meera Kapoor | A+          | AB+            |       95.00 | 2026-08-24 | Suggested |
|       11 |          4 | Kavya Iyer   | AB-         | AB+            |       95.00 | 2026-08-24 | Suggested |
|       12 |          4 | Priya Nair   | A-          | AB+            |       95.00 | 2026-08-24 | Suggested |
|       13 |          4 | Vikram Singh | O-          | AB+            |       95.00 | 2026-08-24 | Suggested |
|       14 |          4 | Sneha Patel  | AB+         | AB+            |       95.00 | 2026-08-24 | Suggested |
|       15 |          4 | Ananya Rao   | A+          | AB+            |       95.00 | 2026-08-24 | Suggested |
|       16 |          4 | Rahul Sharma | O+          | AB+            |       95.00 | 2026-08-24 | Suggested |
|       17 |          5 | Vikram Singh | O-          | O-             |       95.00 | 2026-08-24 | Suggested |
|       18 |          6 | Priya Nair   | A-          | A-             |       95.00 | 2026-08-24 | Suggested |
|       19 |          6 | Vikram Singh | O-          | A-             |       95.00 | 2026-08-24 | Suggested |
+----------+------------+--------------+-------------+----------------+-------------+------------+-----------+
19 rows in set (0.00 sec)
```

TEST BEST DONOR PROCEDURE

```
ysql> CALL FindBestDonor(2);
+----------+---------------+--------------+---------------------+
| Donor_ID | Name          | Blood_Group | Last_Donation_Date |
+----------+---------------+--------------+---------------------+
|       10 | Meera Kapoor | A+           | NULL                |
+----------+---------------+--------------+---------------------+
 row in set (0.00 sec)

uery OK, 0 rows affected (0.01 sec)
```

TEST LOW INVENTORY

```
mysql> SELECT
    ->     h.Hospital_Name,
    ->     bg.Blood_Type,
    ->     i.Available_Units
    -> FROM Inventory i
    -> JOIN Hospital h
    ->     ON i.Hospital_ID = h.Hospital_ID
    -> JOIN Blood_Group bg
    ->     ON i.Blood_Group_ID = bg.Blood_Group_ID
    -> WHERE i.Available_Units < 2;
+------------------+------------+-----------------+
| Hospital_Name    | Blood_Type | Available_Units |
+------------------+------------+-----------------+
| Apollo Hospital  | O+         |               0 |
| Fortis Hospital  | A+         |               0 |
| Manipal Hospital | B+         |               0 |
| Kauvery Hospital | AB+        |               1 |
| Narayana Health  | O-         |               0 |
| Apollo Hospital  | A-         |               0 |
| Fortis Hospital  | AB-        |               1 |
| Manipal Hospital | A+         |               1 |
+------------------+------------+-----------------+
8 rows in set (0.00 sec)
```

TEST EMERGENCY REQUESTS

```
mysql> SELECT
    ->     br.Request_ID,
    ->     p.Name AS Patient_Name,
    ->     h.Hospital_Name,
    ->     bg.Blood_Type,
    ->     br.Units_Required,
    ->     br.Urgency
    -> FROM Blood_Request br
    -> JOIN Patient p
    ->     ON br.Patient_ID = p.Patient_ID
    -> JOIN Hospital h
    ->     ON br.Hospital_ID = h.Hospital_ID
    -> JOIN Blood_Group bg
    ->     ON br.Blood_Group_ID = bg.Blood_Group_ID
    -> WHERE br.Urgency = 'Emergency';
+------------+--------------+------------------+------------+----------------+-----------+
| Request_ID | Patient_Name | Hospital_Name    | Blood_Type | Units_Required | Urgency   |
+------------+--------------+------------------+------------+----------------+-----------+
|          1 | Karthik Rao  | Apollo Hospital  | O+         |              2 | Emergency |
|          3 | Arjun Reddy  | Manipal Hospital | B+         |              2 | Emergency |
|          5 | Ravi Kumar   | Narayana Health  | O-         |              3 | Emergency |
+------------+--------------+------------------+------------+----------------+-----------+
3 rows in set (0.00 sec)

mysql>
```

```
mysql> SELECT
    ->     h.Hospital_Name,
    ->     i.Blood_Group,
    ->     i.Total_Units,
    ->     i.Available_Units,
    ->     i.Reserved_Units
    -> FROM Inventory i
    -> JOIN Hospital h
    ->     ON i.Hospital_ID = h.Hospital_ID
    -> ORDER BY h.Hospital_Name, i.Blood_Group;
+------------------+-------------+-------------+-----------------+----------------+
| Hospital_Name    | Blood_Group | Total_Units | Available_Units | Reserved_Units |
+------------------+-------------+-------------+-----------------+----------------+
| Apollo Hospital  | A-          |           1 |               0 |              0 |
| Apollo Hospital  | O+          |           1 |               0 |              0 |
| Fortis Hospital  | A+          |           1 |               0 |              0 |
| Fortis Hospital  | AB-         |           1 |               1 |              0 |
| Kauvery Hospital | AB+         |           1 |               1 |              0 |
| Manipal Hospital | A+          |           1 |               1 |              0 |
| Manipal Hospital | B+          |           1 |               0 |              0 |
| Narayana Health  | O-          |           1 |               0 |              0 |
+------------------+-------------+-------------+-----------------+----------------+
8 rows in set (0.00 sec)
```

TESTING TRIGGERS

```
+-----------------------+--------------------+
| CheckBloodUnitStatus | UPDATE | blood_unit | BEGIN
    IF NEW.Status = 'Available' AND NEW.Expiry_Date < CURDATE() THEN
        SET NEW.Status = 'Expired';
    END IF;
END

                                              | BEFORE | 2026-08-24 14:07:11.71 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION |
root@localhost | utf8mb4            | utf8mb4_0900_ai_ci    | utf8mb4_0900_ai_ci |
| UpdateInventory      | UPDATE | blood_unit | BEGIN

    IF OLD.Status <> NEW.Status THEN

        IF OLD.Status = 'Available' THEN
            UPDATE Inventory
            SET Available_Units = Available_Units - 1
            WHERE Hospital_ID = OLD.Hospital_ID
              AND Blood_Group = OLD.Blood_Group;
        END IF;

        IF OLD.Status = 'Reserved' THEN
            UPDATE Inventory
            SET Reserved_Units = Reserved_Units - 1
            WHERE Hospital_ID = OLD.Hospital_ID
              AND Blood_Group = OLD.Blood_Group;
        END IF;

        IF NEW.Status = 'Available' THEN
            UPDATE Inventory
            SET Available_Units = Available_Units + 1
            WHERE Hospital_ID = NEW.Hospital_ID
              AND Blood_Group = NEW.Blood_Group;
        END IF;

        IF NEW.Status = 'Reserved' THEN
            UPDATE Inventory
            SET Reserved_Units = Reserved_Units + 1
            WHERE Hospital_ID = NEW.Hospital_ID
              AND Blood_Group = NEW.Blood_Group;
        END IF;

    END IF;

END | AFTER  | 2026-08-24 14:19:28.01 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION | root@localhost | utf8mb4            | utf8mb4_
0900_ai_ci   | utf8mb4_0900_ai_ci |
+-----------------------+--------+----------+-------------------
```

CHECK DONATION CAMPS

```
mysql> SELECT
    ->     d.Donation_ID,
    ->     dn.Name AS Donor_Name,
    ->     dc.Camp_Name,
    ->     dc.Location,
    ->     d.Donation_Date,
    ->     d.Units
    -> FROM Donation d
    -> JOIN Donor dn
    ->     ON d.Donor_ID = dn.Donor_ID
    -> JOIN Donation_Camp dc
    ->     ON d.Camp_ID = dc.Camp_ID;
+-------------+--------------+-------------------------------+-----------+---------------+-------+
| Donation_ID | Donor_Name   | Camp_Name                     | Location  | Donation_Date | Units |
+-------------+--------------+-------------------------------+-----------+---------------+-------+
|           1 | Rahul Sharma | PESU Blood Donation Drive     | Bangalore | 2026-05-20    |     1 |
|           2 | Ananya Rao   | PESU Blood Donation Drive     | Bangalore | 2026-06-15    |     1 |
|           3 | Arjun Kumar  | PESU Blood Donation Drive     | Bangalore | 2026-04-10    |     1 |
|           4 | Sneha Patel  | Life Saver Blood Camp         | Chennai   | 2026-08-10    |     1 |
|           5 | Vikram Singh | Life Saver Blood Camp         | Chennai   | 2026-05-01    |     1 |
|           6 | Priya Nair   | Community Blood Donation Camp | Bangalore | 2026-06-01    |     1 |
|           7 | Aditya Mehta | Community Blood Donation Camp | Bangalore | 2026-07-01    |     1 |
|           8 | Kavya Iyer   | Community Blood Donation Camp | Bangalore | 2026-08-12    |     1 |
|           9 | Rohan Das    | Emergency Blood Donation Drive| Chennai   | 2026-03-15    |     1 |
|          10 | Meera Kapoor | Emergency Blood Donation Drive| Chennai   | 2026-08-15    |     1 |
+-------------+--------------+-------------------------------+-----------+---------------+-------+
10 rows in set (0.00 sec)

mysql>
```

TEST TRANSFER

```
mysql> SELECT
    ->     t.Transfer_ID,
    ->     h1.Hospital_Name AS From_Hospital,
    ->     h2.Hospital_Name AS To_Hospital,
    ->     bu.Unit_ID,
    ->     bu.Blood_Group,
    ->     t.Transfer_Date,
    ->     t.Status
    -> FROM Transfer t
    -> JOIN Hospital h1
    ->     ON t.From_Hospital = h1.Hospital_ID
    -> JOIN Hospital h2
    ->     ON t.To_Hospital = h2.Hospital_ID
    -> JOIN Blood_Unit bu
    ->     ON t.Unit_ID = bu.Unit_ID;
Empty set (0.01 sec)

mysql> INSERT INTO Transfer
    -> (From_Hospital, To_Hospital, Unit_ID, Transfer_Date, Status)
    -> VALUES
    -> (3, 1, 8, '2026-08-24', 'Completed'),
    -> (2, 4, 7, '2026-08-24', 'Requested');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT
    ->     t.Transfer_ID,
    ->     h1.Hospital_Name AS From_Hospital,
    ->     h2.Hospital_Name AS To_Hospital,
    ->     bu.Unit_ID,
    ->     bu.Blood_Group,
    ->     t.Transfer_Date,
    ->     t.Status
    -> FROM Transfer t
    -> JOIN Hospital h1
    ->     ON t.From_Hospital = h1.Hospital_ID
    -> JOIN Hospital h2
    ->     ON t.To_Hospital = h2.Hospital_ID
    -> JOIN Blood_Unit bu
    ->     ON t.Unit_ID = bu.Unit_ID;
+-------------+-----------------+-----------------+---------+-------------+---------------+-----------+
| Transfer_ID | From_Hospital   | To_Hospital     | Unit_ID | Blood_Group | Transfer_Date | Status    |
+-------------+-----------------+-----------------+---------+-------------+---------------+-----------+
|           1 | Manipal Hospital| Apollo Hospital |       8 | A+          | 2026-08-24    | Completed |
|           2 | Fortis Hospital | Kauvery Hospital|       7 | AB-         | 2026-08-24    | Requested |
+-------------+-----------------+-----------------+---------+-------------+---------------+-----------+
2 rows in set (0.00 sec)

mysql>
```

6. DML STATEMENTS

STATEMENTS WITH SCREEN SHOTS OF THE TABLE WITH INSERTED VALUES

SCHEMA OVERVIEW

```
mysql> USE blood_management;
Database changed
mysql>
mysql> SELECT table_name, GROUP_CONCAT(column_name ORDER BY ordinal_position SEPARATOR ', ') AS columns_list
    -> FROM information_schema.columns
    -> WHERE table_schema = 'blood_management'
    -> GROUP BY table_name;
+------------------+-----------------------------------------------------------------------------------------------------------------------------+
| TABLE_NAME       | columns_list                                                                                                                |
+------------------+-----------------------------------------------------------------------------------------------------------------------------+
| blood_group      | Blood_Group_ID, Blood_Type, Description                                                                                     |
| blood_request    | Request_ID, Patient_ID, Hospital_ID, Blood_Group, Units_Required, Urgency, Request_Date, Status, Blood_Group_ID             |
| blood_unit       | Unit_ID, Donation_ID, Blood_Group, Collection_Date, Expiry_Date, Status, Hospital_ID, Blood_Group_ID                        |
| compatible_donors | Request_ID, Required_Group, Donor_ID, Name, Donor_Group, Last_Donation_Date                                                 |
| donation         | Donation_ID, Donor_ID, Donation_Date, Units, Screening_Status, Camp_ID                                                      |
| donation_camp    | Camp_ID, Camp_Name, Location, Organizer, Camp_Date, Start_Time, End_Time, Hospital_ID                                       |
| donor            | Donor_ID, Name, DOB, Gender, Blood_Group, Phone, Address, Last_Donation_Date, Blood_Group_ID                                |
| eligibility      | Eligibility_ID, Donor_ID, Eligible, Reason, Review_Date                                                                     |
| hospital         | Hospital_ID, Hospital_Name, Location, Contact                                                                               |
| inventory        | Inventory_ID, Hospital_ID, Blood_Group, Total_Units, Available_Units, Reserved_Units, Last_Updated, Blood_Group_ID          |
| matching         | Match_ID, Request_ID, Donor_ID, Compatibility, Match_Score, Status, Match_Date                                              |
| patient          | Patient_ID, Hospital_ID, Name, Blood_Group, Medical_Status                                                                  |
| transfer         | Transfer_ID, From_Hospital, To_Hospital, Unit_ID, Transfer_Date, Status                                                     |
+------------------+-----------------------------------------------------------------------------------------------------------------------------+
13 rows in set (0.04 sec)

mysql>
```

```
13 rows in set (0.04 sec)
       SELECT table_name, column_name, referenced_table_name, referenced_column_name
    -> FROM information_schema.key_column_usage
    -> WHERE table_schema = 'blood_management' AND referenced_table_name IS NOT NULL;
+---------------+----------------+-----------------------+-------------------------+
| TABLE_NAME    | COLUMN_NAME    | REFERENCED_TABLE_NAME | REFERENCED_COLUMN_NAME  |
+---------------+----------------+-----------------------+-------------------------+
| blood_request | Patient_ID     | patient               | Patient_ID              |
| blood_request | Hospital_ID    | hospital              | Hospital_ID             |
| blood_request | Blood_Group_ID | blood_group           | Blood_Group_ID          |
| blood_unit    | Donation_ID    | donation              | Donation_ID             |
| blood_unit    | Hospital_ID    | hospital              | Hospital_ID             |
| blood_unit    | Blood_Group_ID | blood_group           | Blood_Group_ID          |
| donation      | Donor_ID       | donor                 | Donor_ID                |
| donation      | Camp_ID        | donation_camp         | Camp_ID                 |
| donation_camp | Hospital_ID    | hospital              | Hospital_ID             |
| donor         | Blood_Group_ID | blood_group           | Blood_Group_ID          |
| eligibility   | Donor_ID       | donor                 | Donor_ID                |
| inventory     | Blood_Group_ID | blood_group           | Blood_Group_ID          |
| inventory     | Hospital_ID    | hospital              | Hospital_ID             |
| matching      | Donor_ID       | donor                 | Donor_ID                |
| matching      | Request_ID     | blood_request         | Request_ID              |
| patient       | Hospital_ID    | hospital              | Hospital_ID             |
| transfer      | From_Hospital  | hospital              | Hospital_ID             |
| transfer      | To_Hospital    | hospital              | Hospital_ID             |
| transfer      | Unit_ID        | blood_unit            | Unit_ID                 |
+---------------+----------------+-----------------------+-------------------------+
19 rows in set (0.04 sec)

mysql>
```

```
mysql> SELECT COUNT(*) FROM donor;
+----------+
| COUNT(*) |
+----------+
|       10 |
+----------+
1 row in set (0.14 sec)
```

INSERT OPERATIONS

HOSPITALS

```
mysql> INSERT INTO hospital (Hospital_ID, Hospital_Name, Location, Contact) VALUES
    -> (101, 'Apollo Test Hospital', 'Chennai', '044-5550101'),
    -> (102, 'Sunrise Test Hospital', 'Bengaluru', '080-5550102');
Query OK, 2 rows affected (0.05 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM hospital WHERE Hospital_ID >= 101;   -- 2 rows
+-------------+-----------------------+------------+--------------+
| Hospital_ID | Hospital_Name         | Location   | Contact      |
+-------------+-----------------------+------------+--------------+
|         101 | Apollo Test Hospital  | Chennai    | 044-5550101 |
|         102 | Sunrise Test Hospital | Bengaluru | 080-5550102 |
+-------------+-----------------------+------------+--------------+
2 rows in set (0.00 sec)

mysql>
```

DONATION CAMP

```
mysql> INSERT INTO donation_camp (Camp_ID, Camp_Name, Location, Organizer, Camp_Date, Start_Time, End_Time, Hospital_ID) VALUES
    -> (101, 'Test Blood Camp', 'Town Hall, Chennai', 'Rotary Club', DATE_SUB(CURDATE(), INTERVAL 10 DAY), '09:00:00', '15:00:00', 101);
Query OK, 1 row affected (0.03 sec)

mysql> SELECT * FROM donation_camp WHERE Camp_ID = 101;   -- 1 row
+---------+----------------+-------------------+------------+------------+------------+----------+-------------+
| Camp_ID | Camp_Name      | Location          | Organizer  | Camp_Date  | Start_Time | End_Time | Hospital_ID |
+---------+----------------+-------------------+------------+------------+------------+----------+-------------+
|     101 | Test Blood Camp | Town Hall, Chennai | Rotary Club | 2026-09-22 | 09:00:00   | 15:00:00 |         101 |
+---------+----------------+-------------------+------------+------------+------------+----------+-------------+
1 row in set (0.00 sec)
```

PATIENTS

```
mysql> INSERT INTO patient (Patient_ID, Hospital_ID, Name, Blood_Group, Medical_Status) VALUES
    -> (101, 101, 'Ravi Shankar', 'O+', 'Critical'),
    -> (102, 102, 'Sita Devi', 'A+', 'Stable');
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM patient WHERE Patient_ID >= 101;     -- 2 rows
+------------+-------------+--------------+-------------+----------------+
| Patient_ID | Hospital_ID | Name         | Blood_Group | Medical_Status |
+------------+-------------+--------------+-------------+----------------+
|        101 |         101 | Ravi Shankar | O+          | Critical       |
|        102 |         102 | Sita Devi    | A+          | Stable         |
+------------+-------------+--------------+-------------+----------------+
2 rows in set (0.00 sec)
```

DONORS

```
mysql> INSERT INTO donor (Donor_ID, Name, DOB, Gender, Blood_Group, Phone, Address, Last_Donation_Date, Blood_Group_ID) VALUES
    -> (101, 'Arjun Kumar', '1990-04-12', 'M', 'O+', '9800000101', 'Adyar, Chennai', NULL,
    ->    (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'O+')),
    -> (102, 'Priya Nair', '1994-08-23', 'F', 'A+', '9800000102', 'Indiranagar, Bengaluru', NULL,
    ->    (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'A+')),
    -> (103, 'Test Donor', '2000-01-01', 'M', 'O+', '9800000103', NULL, NULL,
    ->    (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'O+'));  -- deleted in Part 3
Query OK, 3 rows affected (0.04 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT Donor_ID, Name, Blood_Group, Blood_Group_ID FROM donor WHERE Donor_ID >= 101;   -- 3 rows, no NULL Blood_Group_ID
+----------+-------------+-------------+---------------+
| Donor_ID | Name        | Blood_Group | Blood_Group_ID |
+----------+-------------+-------------+---------------+
|      101 | Arjun Kumar | O+          |             7 |
|      102 | Priya Nair  | A+          |             1 |
|      103 | Test Donor  | O+          |             7 |
+----------+-------------+-------------+---------------+
3 rows in set (0.00 sec)
```

ELIGIBILITY

```
mysql> INSERT INTO eligibility (Donor_ID, Eligible, Reason, Review_Date) VALUES
    -> (101, 'Yes', 'Healthy, cleared to donate', CURDATE()),
    -> (102, 'No', 'Low hemoglobin', CURDATE());
ERROR 1366 (HY000): Incorrect integer value: 'Yes' for column 'Eligible' at row 1
mysql> SELECT * FROM eligibility WHERE Donor_ID >= 101;   -- 2 rows
Empty set (0.02 sec)
```

DONATIONS

```
mysql> INSERT INTO donation (Donation_ID, Donor_ID, Donation_Date, Units, Screening_Status, Camp_ID) VALUES
    -> (101, 101, DATE_SUB(CURDATE(), INTERVAL 10 DAY), 1, 'Passed', 101),
    -> (102, 102, DATE_SUB(CURDATE(), INTERVAL 5 DAY), 1, 'Passed', NULL);
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM donation WHERE Donation_ID >= 101;   -- 2 rows
+-------------+----------+---------------+-------+-----------------+----------+
| Donation_ID | Donor_ID | Donation_Date | Units | Screening_Status | Camp_ID |
+-------------+----------+---------------+-------+-----------------+----------+
|         101 |      101 | 2026-09-22    |     1 | Passed          |      101 |
|         102 |      102 | 2026-09-27    |     1 | Passed          |     NULL |
+-------------+----------+---------------+-------+-----------------+----------+
2 rows in set (0.00 sec)
```

INVENTORY ROWS

```
mysql> INSERT INTO inventory (Inventory_ID, Hospital_ID, Blood_Group, Total_Units, Available_Units, Reserved_Units, Blood_Group_ID) VALUES
    -> (101, 101, 'O+', 1, 1, 0, (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'O+')),
    -> (102, 102, 'A+', 1, 1, 0, (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'A+'));
Query OK, 2 rows affected (0.02 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM inventory WHERE Inventory_ID >= 101;
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+----------------+
| Inventory_ID | Hospital_ID | Blood_Group | Total_Units | Available_Units | Reserved_Units | Last_Updated        | Blood_Group_ID |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+----------------+
|          101 |         101 | O+          |           1 |               1 |              0 | 2026-10-02 17:48:14 |              7 |
|          102 |         102 | A+          |           1 |               1 |              0 | 2026-10-02 17:48:14 |              1 |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+----------------+
2 rows in set (0.00 sec)
```

BLOOD UNITS

```
mysql> INSERT INTO blood_unit (Unit_ID, Donation_ID, Blood_Group, Collection_Date, Expiry_Date, Status, Hospital_ID, Blood_Group_ID) VALUES
    -> (101, 101, 'O+', DATE_SUB(CURDATE(), INTERVAL 10 DAY), DATE_ADD(CURDATE(), INTERVAL 32 DAY), 'Available', 101,
    ->     (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'O+')),
    -> (102, 102, 'A+', DATE_SUB(CURDATE(), INTERVAL 5 DAY), DATE_ADD(CURDATE(), INTERVAL 37 DAY), 'Available', 102,
    ->     (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'A+'));
Query OK, 2 rows affected (0.05 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT Unit_ID, Blood_Group, Status FROM blood_unit WHERE Unit_ID >= 101;   -- 2 rows, Available
+---------+-------------+-----------+
| Unit_ID | Blood_Group | Status    |
+---------+-------------+-----------+
|     101 | O+          | Available |
|     102 | A+          | Available |
+---------+-------------+-----------+
2 rows in set (0.00 sec)

mysql> SELECT Inventory_ID, Total_Units, Available_Units, Reserved_Units FROM inventory WHERE Inventory_ID >= 101;
+--------------+-------------+-----------------+----------------+
| Inventory_ID | Total_Units | Available_Units | Reserved_Units |
+--------------+-------------+-----------------+----------------+
|          101 |           1 |               1 |              0 |
|          102 |           1 |               1 |              0 |
+--------------+-------------+-----------------+----------------+
2 rows in set (0.00 sec)
```

BLOOD REQUESTS

```
mysql> INSERT INTO blood_request (Request_ID, Patient_ID, Hospital_ID, Blood_Group, Units_Required, Urgency, Request_Date, Status, Blood_Group_ID) VALUES
    -> (101, 101, 101, 'O+', 1, 'Critical', CURDATE(), 'Pending', (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'O+')),
    -> (102, 102, 102, 'A+', 1, 'Low', CURDATE(), 'Pending', (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'A+'));
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT Request_ID, Blood_Group, Urgency, Status FROM blood_request WHERE Request_ID >= 101;   -- 2 rows
+------------+-------------+----------+---------+
| Request_ID | Blood_Group | Urgency  | Status  |
+------------+-------------+----------+---------+
|        101 | O+          | Critical | Pending |
|        102 | A+          | Low      | Pending |
+------------+-------------+----------+---------+
2 rows in set (0.00 sec)
```

MATCHING

```
mysql> INSERT INTO matching (Match_ID, Request_ID, Donor_ID, Compatibility, Match_Score, Status, Match_Date) VALUES
    -> (101, 101, 101, 'Exact', 100, 'Proposed', CURDATE()),
    -> (102, 102, 102, 'Exact', 95, 'Proposed', CURDATE());
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM matching WHERE Match_ID >= 101;   -- 2 rows
+----------+------------+----------+---------------+-------------+----------+------------+
| Match_ID | Request_ID | Donor_ID | Compatibility | Match_Score | Status   | Match_Date |
+----------+------------+----------+---------------+-------------+----------+------------+
|      101 |        101 |      101 | Exact         |      100.00 | Proposed | 2026-10-02 |
|      102 |        102 |      102 | Exact         |       95.00 | Proposed | 2026-10-02 |
+----------+------------+----------+---------------+-------------+----------+------------+
2 rows in set (0.00 sec)
```

TRANSFER

```
mysql> INSERT INTO transfer (Transfer_ID, From_Hospital, To_Hospital, Unit_ID, Transfer_Date, Status) VALUES
    -> (101, 101, 102, 101, CURDATE(), 'Pending');
Query OK, 1 row affected (0.03 sec)

mysql> SELECT * FROM transfer WHERE Transfer_ID = 101;   -- 1 row
+-------------+---------------+-------------+---------+---------------+---------+
| Transfer_ID | From_Hospital | To_Hospital | Unit_ID | Transfer_Date | Status  |
+-------------+---------------+-------------+---------+---------------+---------+
|         101 |           101 |         102 |     101 | 2026-10-02    | Pending |
+-------------+---------------+-------------+---------+---------------+---------+
1 row in set (0.00 sec)
```

UPDATE OPERATIONS

Change a donor's contact details

```
mysql> -- 2.1 Change a donor's contact details
mysql> UPDATE donor SET Phone = '9811111111', Address = 'Besant Nagar, Chennai' WHERE Donor_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql> SELECT Donor_ID, Phone, Address FROM donor WHERE Donor_ID = 101;
+----------+------------+-----------------------+
| Donor_ID | Phone      | Address               |
+----------+------------+-----------------------+
|      101 | 9811111111 | Besant Nagar, Chennai |
+----------+------------+-----------------------+
1 row in set (0.00 sec)
```

Update a hospital contact

```
mysql> -- 2.2 Update a hospital contact
mysql> UPDATE hospital SET Contact = '044-5559999' WHERE Hospital_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

Re-test: donor 102 is now eligible

```
mysql> -- 2.3 Re-test: donor 102 is now eligible
mysql> UPDATE eligibility SET Eligible = 'Yes', Reason = 'Hemoglobin recovered', Review_Date = CURDATE
() WHERE Donor_ID = 102;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

Escalate a request's urgency

```
mysql> -- 2.4 Escalate a request's urgency
mysql> UPDATE blood_request SET Urgency = 'High' WHERE Request_ID = 102;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

Reserve a unit (the inventory trigger should fire)

```
mysql> -- 2.5 Reserve a unit (the inventory trigger should fire)
mysql> UPDATE blood_unit SET Status = 'Reserved' WHERE Unit_ID = 101;
Query OK, 0 rows affected (0.01 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

```
mysql> SELECT Hospital_ID, Blood_Group, Available_Units, Reserved_Units FROM inventory WHERE Inventory
_ID = 101;
+-------------+-------------+-----------------+----------------+
| Hospital_ID | Blood_Group | Available_Units | Reserved_Units |
+-------------+-------------+-----------------+----------------+
|         101 | O+          |               0 |              1 |
+-------------+-------------+-----------------+----------------+
1 row in set (0.00 sec)

mysql> -- Available drops by 1 and Reserved rises by 1
```

Confirm the match and mark the request matched

```
mysql> -- 2.6 Confirm the match and mark the request matched
mysql> UPDATE matching SET Status = 'Confirmed' WHERE Match_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql> UPDATE blood_request SET Status = 'Matched' WHERE Request_ID = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Record a donor's latest donation

```
mysql> -- 2.7 Record a donor's latest donation
mysql> UPDATE donor SET Last_Donation_Date = (SELECT Donation_Date FROM donation WHERE Donation_ID = 1
01) WHERE Donor_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

Complete the transfer

```
mysql> -- 2.8 Complete the transfer
mysql> UPDATE transfer SET Status = 'Completed' WHERE Transfer_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

Multi-table UPDATE (JOIN): mark requests Fulfilled when their match is Confirmed

```
mysql> UPDATE blood_request r JOIN matching m ON m.Request_ID = r.Request_ID
    -> SET r.Status = 'Fulfilled'
    -> WHERE m.Status = 'Confirmed' AND r.Request_ID >= 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT Request_ID, Status FROM blood_request WHERE Request_ID >= 101;   -- 101 Fulfilled, 102 P
ending
+------------+-----------+
| Request_ID | Status    |
+------------+-----------+
|        101 | Fulfilled |
|        102 | Pending   |
+------------+-----------+
2 rows in set (0.00 sec)
```

DELETE OPERATIONS

Remove a transfer

```
mysql> -- 3.1 Remove a transfer
mysql> DELETE FROM transfer WHERE Transfer_ID = 101;                    -- 1 row
Query OK, 1 row affected (0.01 sec)
```

Remove an unconfirmed match

```
mysql> -- 3.2 Remove an unconfirmed match
mysql> DELETE FROM matching WHERE Match_ID = 102;                       -- 1 row
Query OK, 1 row affected (0.01 sec)
```

Remove a request

```
mysql> -- 3.3 Remove a request
mysql> DELETE FROM blood_request WHERE Request_ID = 102;                    -- 1 row
Query OK, 1 row affected (0.01 sec)
```

Remove a donor with no donations (safe)

```
mysql> DELETE FROM eligibility WHERE Donor_ID = 103;                        -- 0 rows (none inserted)
Query OK, 0 rows affected (0.00 sec)

mysql> DELETE FROM donor WHERE Donor_ID = 103
    ->   AND NOT EXISTS (SELECT 1 FROM donation WHERE Donor_ID = 103);      -- 1 row
Query OK, 1 row affected (0.00 sec)
```

Remove old matching rows for a donor (optional cleanup pattern)

```
mysql> -- 3.5 Remove old matching rows for a donor (optional cleanup pattern)
mysql> DELETE FROM matching WHERE Status = 'Rejected' AND Match_ID >= 101;  -- 0 rows is fine
Query OK, 0 rows affected (0.00 sec)
```

SELECT OPERATIONS

Row counts for your test rows

```
mysql> -- 4.1 Row counts for your test rows
mysql> SELECT 'hospital' AS tbl, COUNT(*) AS n FROM hospital WHERE Hospital_ID >= 101
    -> UNION ALL SELECT 'donor', COUNT(*) FROM donor WHERE Donor_ID >= 101
    -> UNION ALL SELECT 'donation', COUNT(*) FROM donation WHERE Donation_ID >= 101
    -> UNION ALL SELECT 'blood_unit', COUNT(*) FROM blood_unit WHERE Unit_ID >= 101
    -> UNION ALL SELECT 'blood_request', COUNT(*) FROM blood_request WHERE Request_ID >= 101;
+---------------+---+
| tbl           | n |
+---------------+---+
| hospital      | 2 |
| donor         | 2 |
| donation      | 2 |
| blood_unit    | 2 |
| blood_request | 1 |
+---------------+---+
5 rows in set (0.00 sec)

mysql> -- 2, 2, 2, 2, 1
```

Units with blood group name

```
mysql> -- 4.2 Units with blood group name
mysql> SELECT u.Unit_ID, bg.Blood_Type, h.Hospital_Name, u.Expiry_Date, u.Status
    -> FROM blood_unit u
    -> JOIN blood_group bg ON bg.Blood_Group_ID = u.Blood_Group_ID
    -> JOIN hospital h ON h.Hospital_ID = u.Hospital_ID
    -> WHERE u.Unit_ID >= 101;
+---------+------------+----------------------+-------------+-----------+
| Unit_ID | Blood_Type | Hospital_Name        | Expiry_Date | Status    |
+---------+------------+----------------------+-------------+-----------+
|     101 | O+         | Apollo Test Hospital | 2026-11-03  | Reserved  |
|     102 | A+         | Sunrise Test Hospital| 2026-11-08  | Available |
+---------+------------+----------------------+-------------+-----------+
2 rows in set (0.00 sec)
```

Inventory

```
mysql> -- 4.3 Inventory
mysql> SELECT h.Hospital_Name, i.Blood_Group, i.Total_Units, i.Available_Units, i.Reserved_Units
    -> FROM inventory i JOIN hospital h ON h.Hospital_ID = i.Hospital_ID
    -> WHERE i.Inventory_ID >= 101;
+----------------------+-------------+-------------+-----------------+----------------+
| Hospital_Name        | Blood_Group | Total_Units | Available_Units | Reserved_Units |
+----------------------+-------------+-------------+-----------------+----------------+
| Apollo Test Hospital | O+          |           1 |               0 |              1 |
| Sunrise Test Hospital| A+          |           1 |               1 |              0 |
+----------------------+-------------+-------------+-----------------+----------------+
2 rows in set (0.00 sec)
```

Requests with patient and hospital

```
mysql> SELECT r.Request_ID, p.Name AS patient, h.Hospital_Name, r.Blood_Group, r.Urgency, r.Status
    -> FROM blood_request r
    -> JOIN patient p ON p.Patient_ID = r.Patient_ID
    -> JOIN hospital h ON h.Hospital_ID = r.Hospital_ID
    -> WHERE r.Request_ID >= 101;
+------------+--------------+----------------------+-------------+----------+-----------+
| Request_ID | patient      | Hospital_Name        | Blood_Group | Urgency  | Status    |
+------------+--------------+----------------------+-------------+----------+-----------+
|        101 | Ravi Shankar | Apollo Test Hospital | O+          | Critical | Fulfilled |
+------------+--------------+----------------------+-------------+----------+-----------+
1 row in set (0.00 sec)
```

Donor donation summary

```
mysql> -- 4.5 Donor donation summary
mysql> SELECT d.Donor_ID, d.Name, COUNT(dn.Donation_ID) AS donations, MAX(dn.Donation_Date) AS last_do
nation
    -> FROM donor d LEFT JOIN donation dn ON dn.Donor_ID = d.Donor_ID
    -> GROUP BY d.Donor_ID, d.Name
    -> ORDER BY donations DESC;
+----------+--------------+-----------+---------------+
| Donor_ID | Name         | donations | last_donation |
+----------+--------------+-----------+---------------+
|        1 | Rahul Sharma |         1 | 2026-05-20    |
|        2 | Ananya Rao   |         1 | 2026-06-15    |
|        3 | Arjun Kumar  |         1 | 2026-04-10    |
|        4 | Sneha Patel  |         1 | 2026-08-10    |
|        5 | Vikram Singh |         1 | 2026-05-01    |
|        6 | Priya Nair   |         1 | 2026-06-01    |
|        7 | Aditya Mehta |         1 | 2026-07-01    |
|        8 | Kavya Iyer   |         1 | 2026-08-12    |
|        9 | Rohan Das    |         1 | 2026-03-15    |
|       10 | Meera Kapoor |         1 | 2026-08-15    |
|      101 | Arjun Kumar  |         1 | 2026-09-22    |
|      102 | Priya Nair   |         1 | 2026-09-27    |
+----------+--------------+-----------+---------------+
12 rows in set (0.00 sec)
```

```
mysql> SET SQL_SAFE_UPDATES = 1;
Query OK, 0 rows affected (0.00 sec)
```

7. QUERIES

7.1 SIMPLE QUERY WITH GROUP BY, AGRREGATE

Query 1: Row counts of the inserted test rows (COUNT with UNION ALL)

```
mysql> -- 4.1 Row counts for your test rows
mysql> SELECT 'hospital' AS tbl, COUNT(*) AS n FROM hospital WHERE Hospital_ID >= 101
    -> UNION ALL SELECT 'donor', COUNT(*) FROM donor WHERE Donor_ID >= 101
    -> UNION ALL SELECT 'donation', COUNT(*) FROM donation WHERE Donation_ID >= 101
    -> UNION ALL SELECT 'blood_unit', COUNT(*) FROM blood_unit WHERE Unit_ID >= 101
    -> UNION ALL SELECT 'blood_request', COUNT(*) FROM blood_request WHERE Request_ID >= 101;
+---------------+---+
| tbl           | n |
+---------------+---+
| hospital      | 2 |
| donor         | 2 |
| donation      | 2 |
| blood_unit    | 2 |
| blood_request | 1 |
+---------------+---+
5 rows in set (0.00 sec)

mysql> -- 2, 2, 2, 2, 1
```

Query 2: Donation count and latest donation date per donor (COUNT, MAX, GROUP BY)

```
mysql> -- 4.5 Donor donation summary
mysql> SELECT d.Donor_ID, d.Name, COUNT(dn.Donation_ID) AS donations, MAX(dn.Donation_Date) AS last_do
nation
    -> FROM donor d LEFT JOIN donation dn ON dn.Donor_ID = d.Donor_ID
    -> GROUP BY d.Donor_ID, d.Name
    -> ORDER BY donations DESC;
+----------+--------------+-----------+---------------+
| Donor_ID | Name         | donations | last_donation |
+----------+--------------+-----------+---------------+
|        1 | Rahul Sharma |         1 | 2026-05-20    |
|        2 | Ananya Rao   |         1 | 2026-06-15    |
|        3 | Arjun Kumar  |         1 | 2026-04-10    |
|        4 | Sneha Patel  |         1 | 2026-08-10    |
|        5 | Vikram Singh |         1 | 2026-05-01    |
|        6 | Priya Nair   |         1 | 2026-06-01    |
|        7 | Aditya Mehta |         1 | 2026-07-01    |
|        8 | Kavya Iyer   |         1 | 2026-08-12    |
|        9 | Rohan Das    |         1 | 2026-03-15    |
|       10 | Meera Kapoor |         1 | 2026-08-15    |
|      101 | Arjun Kumar  |         1 | 2026-09-22    |
|      102 | Priya Nair   |         1 | 2026-09-27    |
+----------+--------------+-----------+---------------+
12 rows in set (0.00 sec)
```

Query 3: Available units per hospital and blood group (COUNT, GROUP BY)

```
mysql> SELECT
    ->     h.Hospital_Name,
    ->     b.Blood_Group,
    ->     COUNT(b.Unit_ID) AS Available_Units
    -> FROM Blood_Unit b
    -> JOIN Hospital h
    -> ON b.Hospital_ID = h.Hospital_ID
    -> WHERE b.Status = 'Available'
    -> GROUP BY h.Hospital_Name, b.Blood_Group;
+------------------+-------------+-----------------+
| Hospital_Name    | Blood_Group | Available_Units |
+------------------+-------------+-----------------+
| Kauvery Hospital | AB+         |               1 |
| Fortis Hospital  | AB-         |               1 |
| Manipal Hospital | A+          |               1 |
+------------------+-------------+-----------------+
3 rows in set (0.01 sec)
```

Query 4: Hospitals with low inventory (GROUP BY with HAVING)

```
mysql> SELECT
    ->     h.Hospital_Name,
    ->     b.Blood_Group,
    ->     COUNT(b.Unit_ID) AS Available_Units
    -> FROM Blood_Unit b
    -> JOIN Hospital h
    -> ON b.Hospital_ID = h.Hospital_ID
    -> WHERE b.Status = 'Available'
    -> GROUP BY h.Hospital_Name, b.Blood_Group
    -> HAVING COUNT(b.Unit_ID) < 2;
+------------------+-------------+-----------------+
| Hospital_Name    | Blood_Group | Available_Units |
+------------------+-------------+-----------------+
| Kauvery Hospital | AB+         |               1 |
| Fortis Hospital  | AB-         |               1 |
| Manipal Hospital | A+          |               1 |
+------------------+-------------+-----------------+
3 rows in set (0.00 sec)
```

Query 5: Total donations per donation camp (COUNT, GROUP BY)

```
mysql> SELECT
    ->     dc.Camp_Name,
    ->     dc.Location,
    ->     dc.Camp_Date,
    ->     h.Hospital_Name,
    ->     COUNT(d.Donation_ID) AS Total_Donations
    -> FROM Donation_Camp dc
    -> JOIN Hospital h
    ->     ON dc.Hospital_ID = h.Hospital_ID
    -> LEFT JOIN Donation d
    ->     ON dc.Camp_ID = d.Camp_ID
    -> GROUP BY
    ->     dc.Camp_ID,
    ->     dc.Camp_Name,
    ->     dc.Location,
    ->     dc.Camp_Date,
    ->     h.Hospital_Name;
+-------------------------------+-----------+------------+------------------+-----------------+
| Camp_Name                     | Location  | Camp_Date  | Hospital_Name    | Total_Donations |
+-------------------------------+-----------+------------+------------------+-----------------+
| PESU Blood Donation Drive     | Bangalore | 2026-09-05 | Fortis Hospital  |               3 |
| Life Saver Blood Camp         | Chennai   | 2026-09-12 | Apollo Hospital  |               2 |
| Community Blood Donation Camp | Bangalore | 2026-09-20 | Narayana Health  |               3 |
| Emergency Blood Donation Drive | Chennai  | 2026-10-03 | Kauvery Hospital |               2 |
+-------------------------------+-----------+------------+------------------+-----------------+
4 rows in set (0.01 sec)
```

7.2 UPDATE OPERATION

Update a donor's contact details

```
mysql> -- 2.1 Change a donor's contact details
mysql> UPDATE donor SET Phone = '9811111111', Address = 'Besant Nagar, Chennai' WHERE Donor_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql> SELECT Donor_ID, Phone, Address FROM donor WHERE Donor_ID = 101;
+----------+------------+----------------------+
| Donor_ID | Phone      | Address              |
+----------+------------+----------------------+
|      101 | 9811111111 | Besant Nagar, Chennai |
+----------+------------+----------------------+
1 row in set (0.00 sec)
```

Update a hospital contact

```
mysql> -- 2.2 Update a hospital contact
mysql> UPDATE hospital SET Contact = '044-5559999' WHERE Hospital_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

Mark donor 102 as eligible

```
mysql> -- 2.3 Re-test: donor 102 is now eligible
mysql> UPDATE eligibility SET Eligible = 'Yes', Reason = 'Hemoglobin recovered', Review_Date = CURDATE
() WHERE Donor_ID = 102;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

Escalate a request's urgency

```
mysql> -- 2.4 Escalate a request's urgency
mysql> UPDATE blood_request SET Urgency = 'High' WHERE Request_ID = 102;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

Reserve a unit (the inventory trigger fires)

```
mysql> -- 2.5 Reserve a unit (the inventory trigger should fire)
mysql> UPDATE blood_unit SET Status = 'Reserved' WHERE Unit_ID = 101;
Query OK, 0 rows affected (0.01 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

```
mysql> SELECT Hospital_ID, Blood_Group, Available_Units, Reserved_Units FROM inventory WHERE Inventory
_ID = 101;
+-------------+-------------+-----------------+----------------+
| Hospital_ID | Blood_Group | Available_Units | Reserved_Units |
+-------------+-------------+-----------------+----------------+
|         101 | O+          |               0 |              1 |
+-------------+-------------+-----------------+----------------+
1 row in set (0.00 sec)

mysql> -- Available drops by 1 and Reserved rises by 1
```

Confirm the match and mark the request as matched

```
mysql> -- 2.6 Confirm the match and mark the request matched
mysql> UPDATE matching SET Status = 'Confirmed' WHERE Match_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql> UPDATE blood_request SET Status = 'Matched' WHERE Request_ID = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Record a donor's latest donation

```
mysql> -- 2.7 Record a donor's latest donation
mysql> UPDATE donor SET Last_Donation_Date = (SELECT Donation_Date FROM donation WHERE Donation_ID = 1
01) WHERE Donor_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

Complete the transfer

```
mysql> -- 2.8 Complete the transfer
mysql> UPDATE transfer SET Status = 'Completed' WHERE Transfer_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

Multi-table UPDATE with JOIN: mark requests Fulfilled when their match is Confirmed

```
mysql> UPDATE blood_request r JOIN matching m ON m.Request_ID = r.Request_ID
    -> SET r.Status = 'Fulfilled'
    -> WHERE m.Status = 'Confirmed' AND r.Request_ID >= 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT Request_ID, Status FROM blood_request WHERE Request_ID >= 101;   -- 101 Fulfilled, 102 P
ending
+------------+-----------+
| Request_ID | Status    |
+------------+-----------+
|        101 | Fulfilled |
|        102 | Pending   |
+------------+-----------+
2 rows in set (0.00 sec)
```

7.3 DELETE OPERATION

Remove a transfer

```
mysql> -- 3.1 Remove a transfer
mysql> DELETE FROM transfer WHERE Transfer_ID = 101;                    -- 1 row
Query OK, 1 row affected (0.01 sec)
```

Remove an unconfirmed match

```
mysql> -- 3.2 Remove an unconfirmed match
mysql> DELETE FROM matching WHERE Match_ID = 102;                       -- 1 row
Query OK, 1 row affected (0.01 sec)
```

Remove a request

```
mysql> -- 3.3 Remove a request
mysql> DELETE FROM blood_request WHERE Request_ID = 102;                    -- 1 row
Query OK, 1 row affected (0.01 sec)
```

Remove a donor with no donations

```
mysql> DELETE FROM eligibility WHERE Donor_ID = 103;                    -- 0 rows (none inserted)
Query OK, 0 rows affected (0.00 sec)

mysql> DELETE FROM donor WHERE Donor_ID = 103
    ->   AND NOT EXISTS (SELECT 1 FROM donation WHERE Donor_ID = 103);      -- 1 row
Query OK, 1 row affected (0.00 sec)
```

Remove old rejected matching rows

```
mysql> -- 3.5 Remove old matching rows for a donor (optional cleanup pattern)
mysql> DELETE FROM matching WHERE Status = 'Rejected' AND Match_ID >= 101;  -- 0 rows is fine
Query OK, 0 rows affected (0.00 sec)
```

7.4 CORRELATED QUERY

Highest-scoring donor for each request (subquery correlated on Request_ID)

```
mysql> SELECT
    ->     m.Request_ID,
    ->     d.Name AS Donor_Name,
    ->     d.Blood_Group,
    ->     m.Match_Score,
    ->     m.Status
    -> FROM Matching m
    -> JOIN Donor d
    ->     ON m.Donor_ID = d.Donor_ID
    -> WHERE m.Match_Score = (
    ->     SELECT MAX(m2.Match_Score)
    ->     FROM Matching m2
    ->     WHERE m2.Request_ID = m.Request_ID
    -> );
+------------+--------------+-------------+-------------+-----------+
| Request_ID | Donor_Name   | Blood_Group | Match_Score | Status    |
+------------+--------------+-------------+-------------+-----------+
|          1 | Vikram Singh | O-          |       95.00 | Suggested |
|          1 | Rahul Sharma | O+          |       95.00 | Suggested |
|          2 | Meera Kapoor | A+          |       95.00 | Suggested |
|          2 | Priya Nair   | A-          |       95.00 | Suggested |
|          2 | Vikram Singh | O-          |       95.00 | Suggested |
|          2 | Ananya Rao   | A+          |       95.00 | Suggested |
|          2 | Rahul Sharma | O+          |       95.00 | Suggested |
|          3 | Vikram Singh | O-          |       95.00 | Suggested |
|          3 | Rahul Sharma | O+          |       95.00 | Suggested |
|          4 | Meera Kapoor | A+          |       95.00 | Suggested |
|          4 | Kavya Iyer   | AB-         |       95.00 | Suggested |
|          4 | Priya Nair   | A-          |       95.00 | Suggested |
|          4 | Vikram Singh | O-          |       95.00 | Suggested |
|          4 | Sneha Patel  | AB+         |       95.00 | Suggested |
|          4 | Ananya Rao   | A+          |       95.00 | Suggested |
|          4 | Rahul Sharma | O+          |       95.00 | Suggested |
|          5 | Vikram Singh | O-          |       95.00 | Suggested |
|          6 | Priya Nair   | A-          |       95.00 | Suggested |
|          6 | Vikram Singh | O-          |       95.00 | Suggested |
+------------+--------------+-------------+-------------+-----------+
19 rows in set (0.00 sec)
```

Delete a donor only if no donation exists (correlated NOT EXISTS)

```
mysql> DELETE FROM eligibility WHERE Donor_ID = 103;                        -- 0 rows (none inserted)
Query OK, 0 rows affected (0.00 sec)

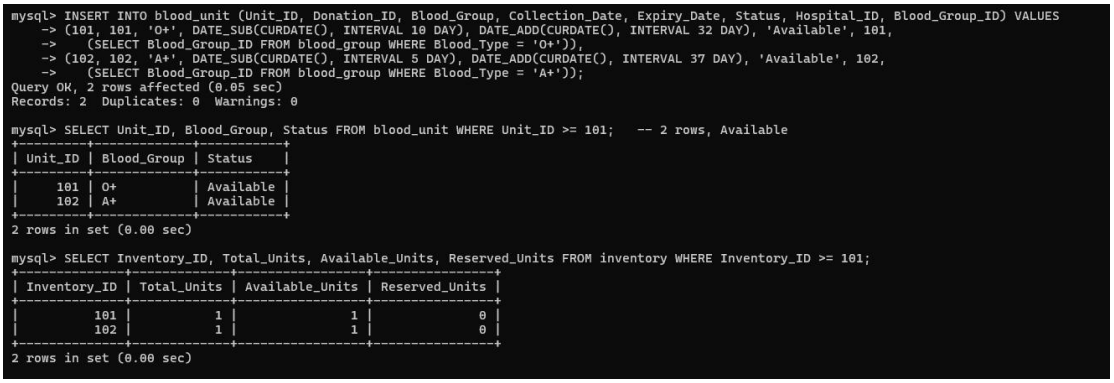
mysql> DELETE FROM donor WHERE Donor_ID = 103
    ->   AND NOT EXISTS (SELECT 1 FROM donation WHERE Donor_ID = 103);      -- 1 row
Query OK, 1 row affected (0.00 sec)
```

7.5 NESTED QUERY

Update a donor's last donation date from a nested SELECT on Donation

```
mysql> -- 2.7 Record a donor's latest donation
mysql> UPDATE donor SET Last_Donation_Date = (SELECT Donation_Date FROM donation WHERE Donation_ID = 1
01) WHERE Donor_ID = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

Insert blood units with the blood group looked up by a nested SELECT



```
mysql> INSERT INTO blood_unit (Unit_ID, Donation_ID, Blood_Group, Collection_Date, Expiry_Date, Status, Hospital_ID, Blood_Group_ID) VALUES
    -> (101, 101, 'O+', DATE_SUB(CURDATE(), INTERVAL 10 DAY), DATE_ADD(CURDATE(), INTERVAL 32 DAY), 'Available', 101,
    ->    (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'O+')),
    -> (102, 102, 'A+', DATE_SUB(CURDATE(), INTERVAL 5 DAY), DATE_ADD(CURDATE(), INTERVAL 37 DAY), 'Available', 102,
    ->    (SELECT Blood_Group_ID FROM blood_group WHERE Blood_Type = 'A+'));
Query OK, 2 rows affected (0.05 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT Unit_ID, Blood_Group, Status FROM blood_unit WHERE Unit_ID >= 101;   -- 2 rows, Available
+---------+-------------+-----------+
| Unit_ID | Blood_Group | Status    |
+---------+-------------+-----------+
|     101 | O+          | Available |
|     102 | A+          | Available |
+---------+-------------+-----------+
2 rows in set (0.00 sec)

mysql> SELECT Inventory_ID, Total_Units, Available_Units, Reserved_Units FROM inventory WHERE Inventory_ID >= 101;
+--------------+-------------+-----------------+----------------+
| Inventory_ID | Total_Units | Available_Units | Reserved_Units |
+--------------+-------------+-----------------+----------------+
|          101 |           1 |               1 |              0 |
|          102 |           1 |               1 |              0 |
+--------------+-------------+-----------------+----------------+
2 rows in set (0.00 sec)
```

Insert matching records from a SELECT on the Compatible_Donors view

```
mysql> INSERT INTO Match_Result
    -> (Request_ID, Donor_ID, Compatibility, Match_Score, Match_Status)
    ->
    -> SELECT
    ->     Request_ID,
    ->     Donor_ID,
    ->     'Compatible',
    ->     95,
    ->     'Suggested'
    -> FROM Compatible_Donors;
Query OK, 19 rows affected (0.01 sec)
Records: 19  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Match_Result;
+----------+------------+----------+---------------+-------------+--------------+
| Match_ID | Request_ID | Donor_ID | Compatibility | Match_Score | Match_Status |
+----------+------------+----------+---------------+-------------+--------------+
|        1 |          1 |        5 | Compatible    |       95.00 | Suggested    |
|        2 |          1 |        1 | Compatible    |       95.00 | Suggested    |
|        3 |          2 |       10 | Compatible    |       95.00 | Suggested    |
|        4 |          2 |        6 | Compatible    |       95.00 | Suggested    |
|        5 |          2 |        5 | Compatible    |       95.00 | Suggested    |
|        6 |          2 |        2 | Compatible    |       95.00 | Suggested    |
|        7 |          2 |        1 | Compatible    |       95.00 | Suggested    |
|        8 |          3 |        5 | Compatible    |       95.00 | Suggested    |
|        9 |          3 |        1 | Compatible    |       95.00 | Suggested    |
|       10 |          4 |       10 | Compatible    |       95.00 | Suggested    |
|       11 |          4 |        8 | Compatible    |       95.00 | Suggested    |
|       12 |          4 |        6 | Compatible    |       95.00 | Suggested    |
|       13 |          4 |        5 | Compatible    |       95.00 | Suggested    |
|       14 |          4 |        4 | Compatible    |       95.00 | Suggested    |
|       15 |          4 |        2 | Compatible    |       95.00 | Suggested    |
|       16 |          4 |        1 | Compatible    |       95.00 | Suggested    |
|       17 |          5 |        5 | Compatible    |       95.00 | Suggested    |
|       18 |          6 |        6 | Compatible    |       95.00 | Suggested    |
|       19 |          6 |        5 | Compatible    |       95.00 | Suggested    |
+----------+------------+----------+---------------+-------------+--------------+
19 rows in set (0.00 sec)
```

8. STORED PROCEDURES, FUCNTIONS AND TRIGGERS

8.1 STORED PROCEDURES OR FUNCTIONS

Stored procedure FindBestDonor and a call for request 1

```
mysql> CREATE PROCEDURE FindBestDonor(IN req INT)
    -> BEGIN
    ->     SELECT
    ->         d.Donor_ID,
    ->         d.Name,
    ->         d.Blood_Group,
    ->         d.Last_Donation_Date
    ->     FROM Compatible_Donors cd
    ->     JOIN Donor d
    ->     ON cd.Donor_ID=d.Donor_ID
    ->     WHERE cd.Request_ID=req
    ->     ORDER BY d.Last_Donation_Date ASC
    ->     LIMIT 1;
    -> END //
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL FindBestDonor(1);
+----------+--------------+-------------+--------------------+
| Donor_ID | Name         | Blood_Group | Last_Donation_Date |
+----------+--------------+-------------+--------------------+
|        5 | Vikram Singh | O-          | 2026-05-01         |
+----------+--------------+-------------+--------------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql>
```

Calling FindBestDonor for request 2

```
ysql> CALL FindBestDonor(2);
----------+--------------+-------------+--------------------+
 Donor_ID | Name         | Blood_Group | Last_Donation_Date |
----------+--------------+-------------+--------------------+
       10 | Meera Kapoor | A+          | NULL               |
----------+--------------+-------------+--------------------+
 row in set (0.00 sec)

uery OK, 0 rows affected (0.01 sec)
```

8.2 TRIGGERS

Dropping the existing UpdateInventory trigger

```
mysql> DROP TRIGGER IF EXISTS UpdateInventory;
Query OK, 0 rows affected, 1 warning (0.01 sec)
```

Creating the UpdateInventory trigger (AFTER UPDATE on Blood_Unit)

```
mysql>
mysql> CREATE TRIGGER UpdateInventory
    -> AFTER UPDATE ON Blood_Unit
    -> FOR EACH ROW
    -> BEGIN
    ->
    ->     IF OLD.Status <> NEW.Status THEN
    ->
    ->         IF OLD.Status = 'Available' THEN
    ->             UPDATE Inventory
    ->             SET Available_Units = Available_Units - 1
    ->             WHERE Hospital_ID = OLD.Hospital_ID
    ->               AND Blood_Group = OLD.Blood_Group;
    ->         END IF;
    ->
    ->         IF OLD.Status = 'Reserved' THEN
    ->             UPDATE Inventory
    ->             SET Reserved_Units = Reserved_Units - 1
    ->             WHERE Hospital_ID = OLD.Hospital_ID
    ->               AND Blood_Group = OLD.Blood_Group;
    ->         END IF;
    ->
    ->         IF NEW.Status = 'Available' THEN
    ->             UPDATE Inventory
    ->             SET Available_Units = Available_Units + 1
    ->             WHERE Hospital_ID = NEW.Hospital_ID
    ->               AND Blood_Group = NEW.Blood_Group;
    ->         END IF;
    ->
    ->         IF NEW.Status = 'Reserved' THEN
    ->             UPDATE Inventory
    ->             SET Reserved_Units = Reserved_Units + 1
    ->             WHERE Hospital_ID = NEW.Hospital_ID
    ->               AND Blood_Group = NEW.Blood_Group;
    ->         END IF;
    ->
    ->     END IF;
    ->
    -> END //
Query OK, 0 rows affected (0.01 sec)
```

Showing the triggers in the database



Testing the trigger: inventory before the update

```
mysql> SELECT * FROM Inventory
    -> WHERE Hospital_ID = 3
    -> AND Blood_Group = 'A+';
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
| Inventory_ID | Hospital_ID | Blood_Group | Total_Units | Available_Units | Reserved_Units | Last_Updated        |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
|            8 |           3 | A+          |           1 |               1 |              0 | 2026-08-24 14:18:03 |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
1 row in set (0.00 sec)
```

Reserving unit 8 (Available to Reserved)

```
mysql> UPDATE Blood_Unit
    -> SET Status = 'Reserved'
    -> WHERE Unit_ID = 8;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Inventory after reserving: available units changed from 1 to 0

```
mysql> SELECT * FROM Inventory
    -> WHERE Hospital_ID = 3
    -> AND Blood_Group = 'A+';
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
| Inventory_ID | Hospital_ID | Blood_Group | Total_Units | Available_Units | Reserved_Units | Last_Updated        |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
|            8 |           3 | A+          |           1 |               0 |              1 | 2026-08-24 14:23:03 |
+--------------+-------------+-------------+-------------+-----------------+----------------+---------------------+
1 row in set (0.00 sec)
```

Reversing the operation (Reserved to Available)

```
mysql> UPDATE Blood_Unit
    -> SET Status = 'Available'
    -> WHERE Unit_ID = 8;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Inventory after reversing: available units changed from 0 to 1

```
mysql> SELECT * FROM Inventory
    -> WHERE Hospital_ID = 3
    -> AND Blood_Group = 'A+';
+--------------+-------------+-------------+-------------+-----------------+---------------+---------------------+
| Inventory_ID | Hospital_ID | Blood_Group | Total_Units | Available_Units | Reserved_Units | Last_Updated        |
+--------------+-------------+-------------+-------------+-----------------+---------------+---------------------+
|            8 |           3 | A+          |           1 |               1 |             0 | 2026-08-24 14:25:58 |
+--------------+-------------+-------------+-------------+-----------------+---------------+---------------------+
1 row in set (0.00 sec)
```

SUMMARY OF QUERIES, PROCEDURES AND TRIGGERS

The table below summarises the screenshots presented in Chapters 7 and 8.

Section	Screenshots used
7.1 Simple query (GROUP BY, aggregate)	Row counts (COUNT with UNION ALL), donations per donor (COUNT, MAX, GROUP BY), available units per hospital (GROUP BY), low inventory (HAVING), donations per camp
7.2 Update	All nine updates, including the trigger-firing reserve and the JOIN update
7.3 Delete	All five deletes
7.4 Correlated query	Highest-scoring donor per request (subquery correlated on Request_ID), and the delete using NOT EXISTS
7.5 Nested query	Last donation date set from a nested SELECT, blood units inserted with a nested SELECT, and matching records inserted from the Compatible_Donors view
8.1 Stored procedures	FindBestDonor created and called for request 1, then called for request 2
8.2 Triggers	Drop and create UpdateInventory, SHOW TRIGGERS, and the test (inventory before, reserve, 1 to 0, reverse, 0 to 1)

9. FRONT END DEVELOPEMNT

The front end is a web application that lets staff manage the blood bank database through a set of screens: an overview dashboard, inventory, requests and matching, donors, donations, eligibility, camps, hospitals, users and an audit trail. The screenshots of each screen are shown below.

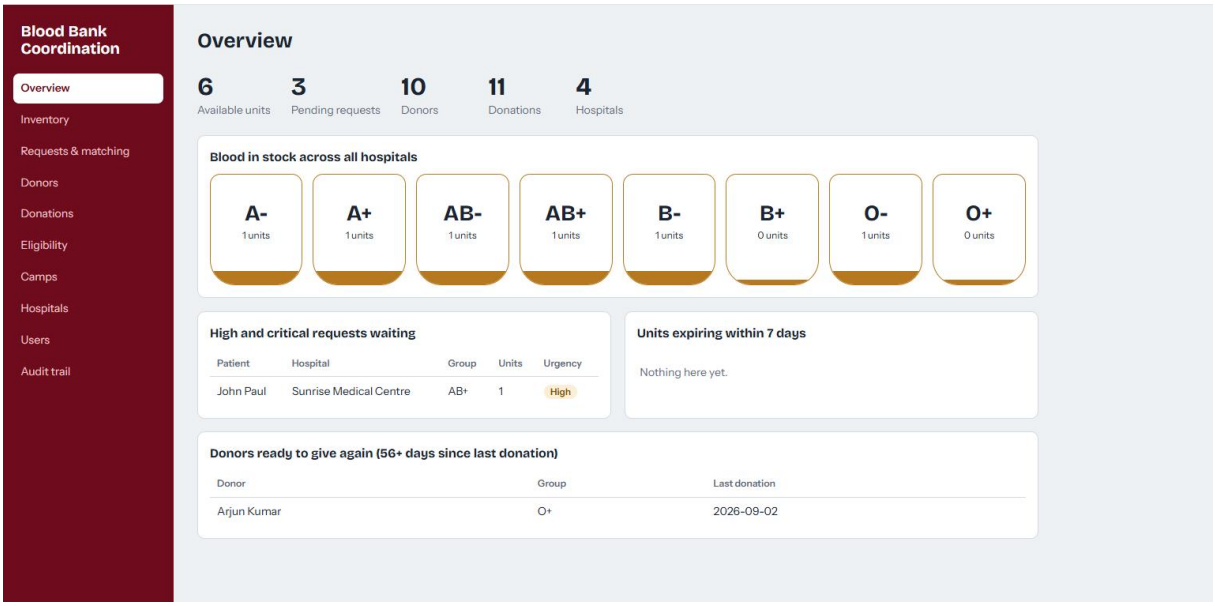


Fig 9.1 Overview dashboard showing stock by blood group, critical requests and donors ready to give again

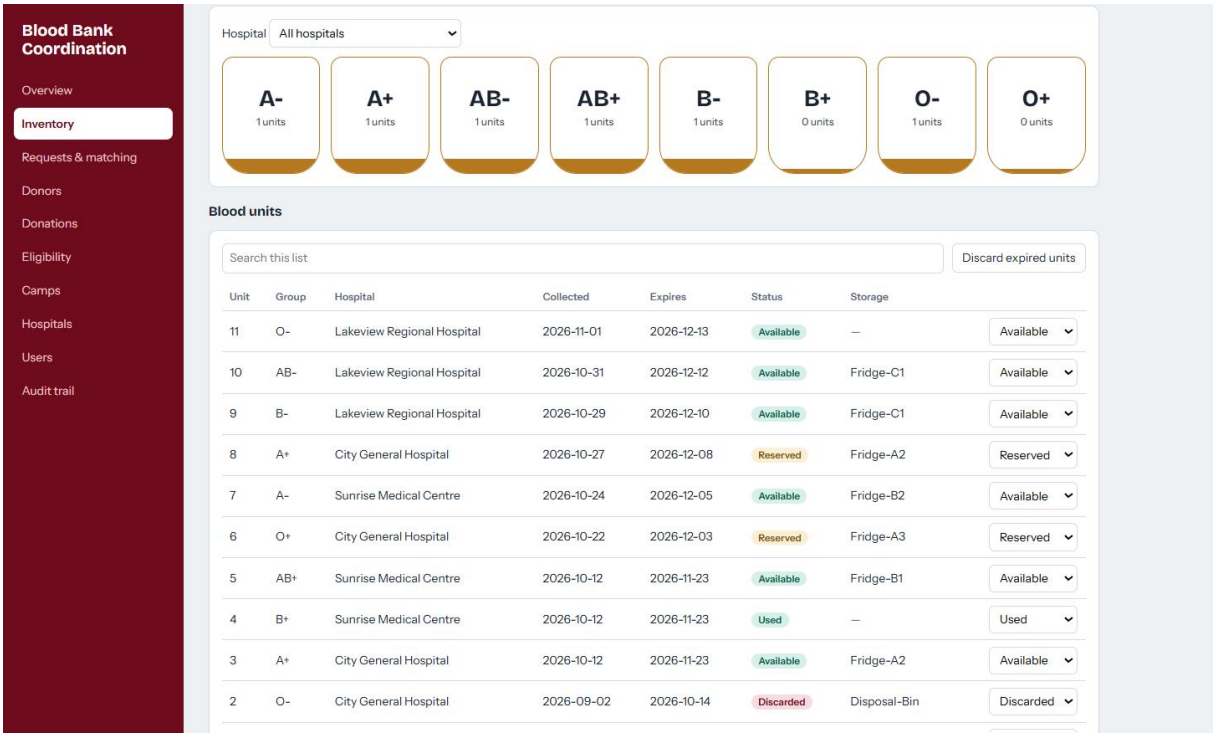


Fig 9.2 Inventory screen with blood units and their status

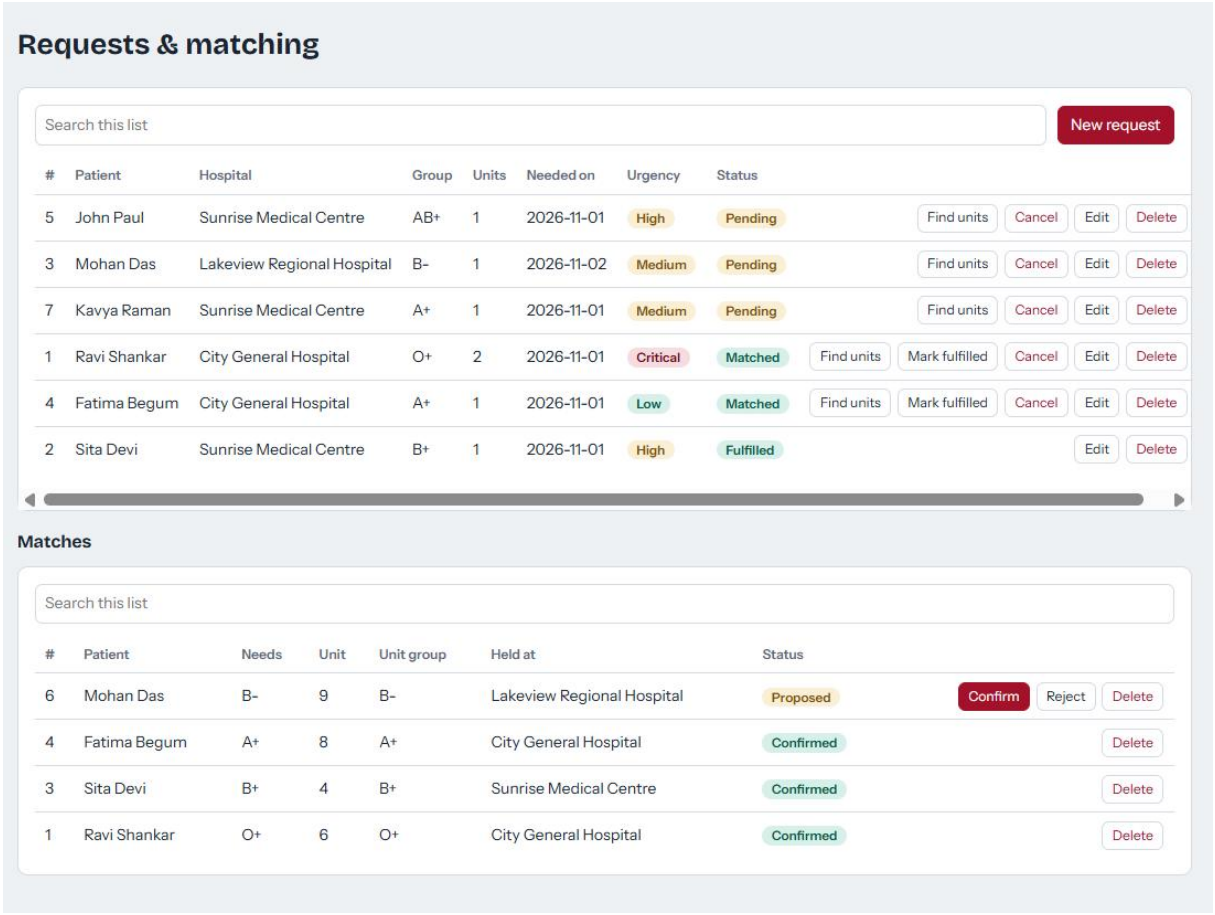


Fig 9.3 Requests and matching screen

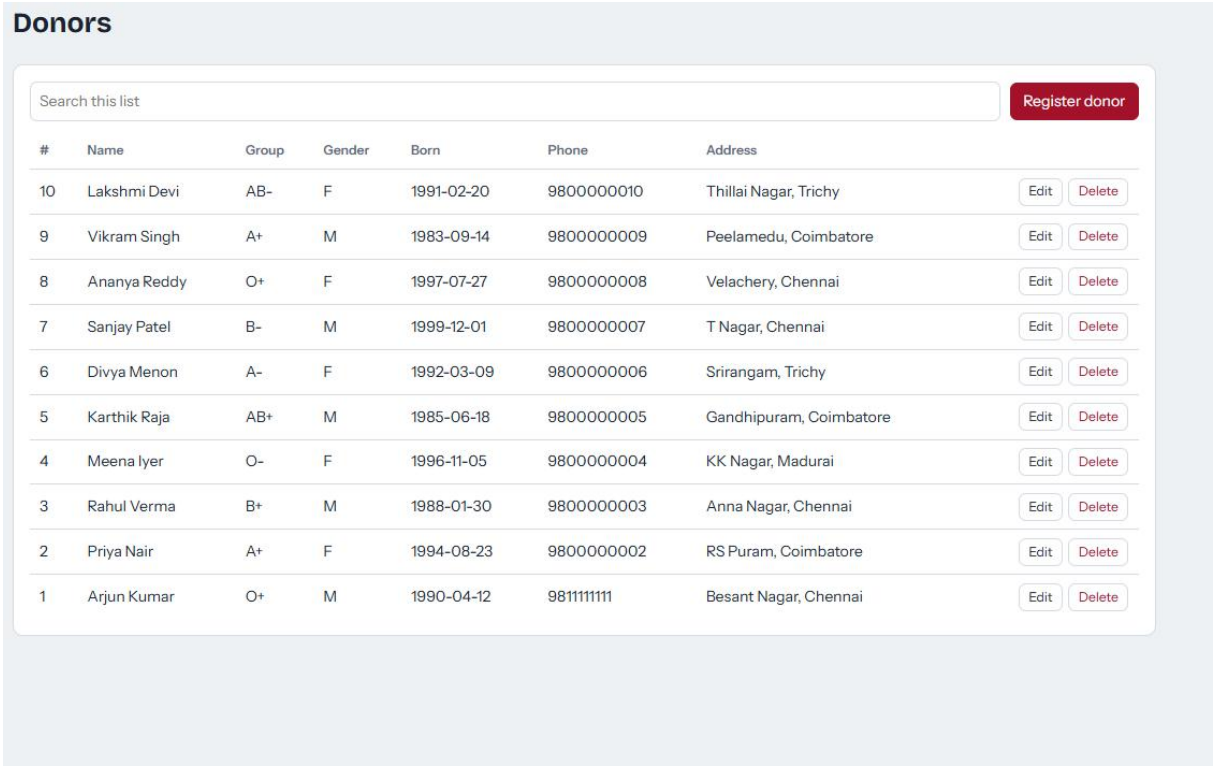


Fig 9.4 Donors screen

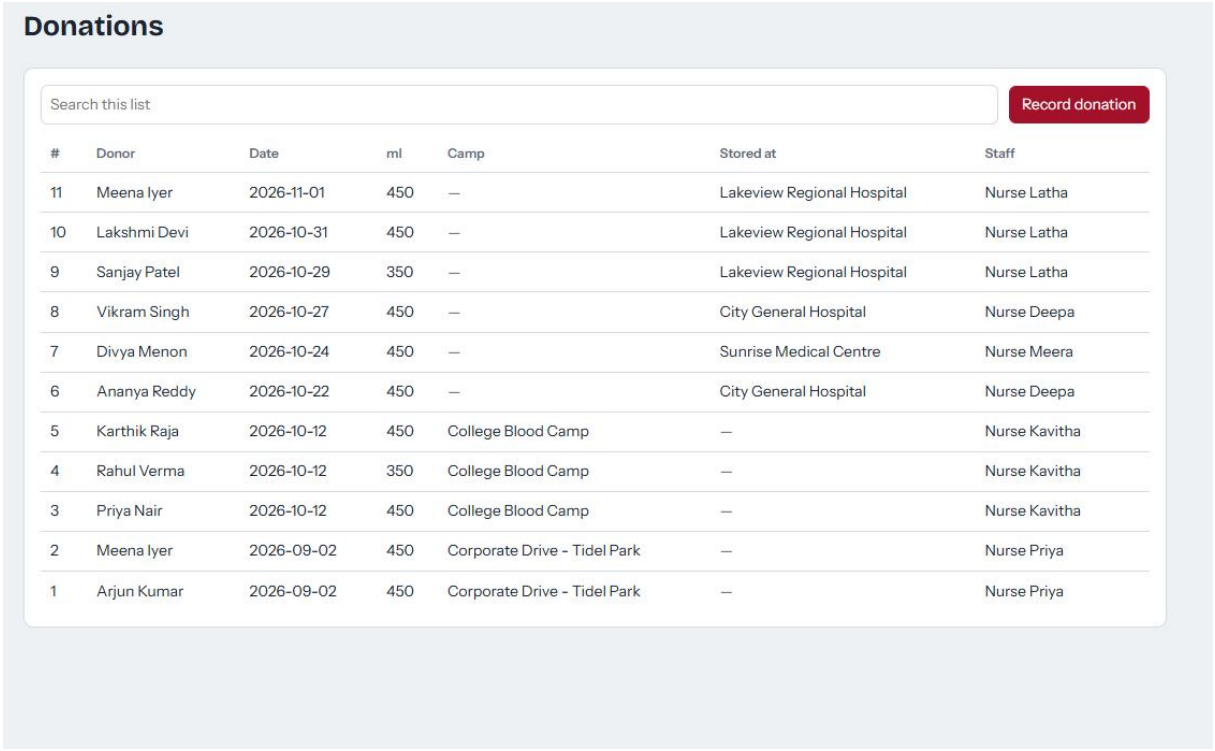


Fig 9.5 Donations screen

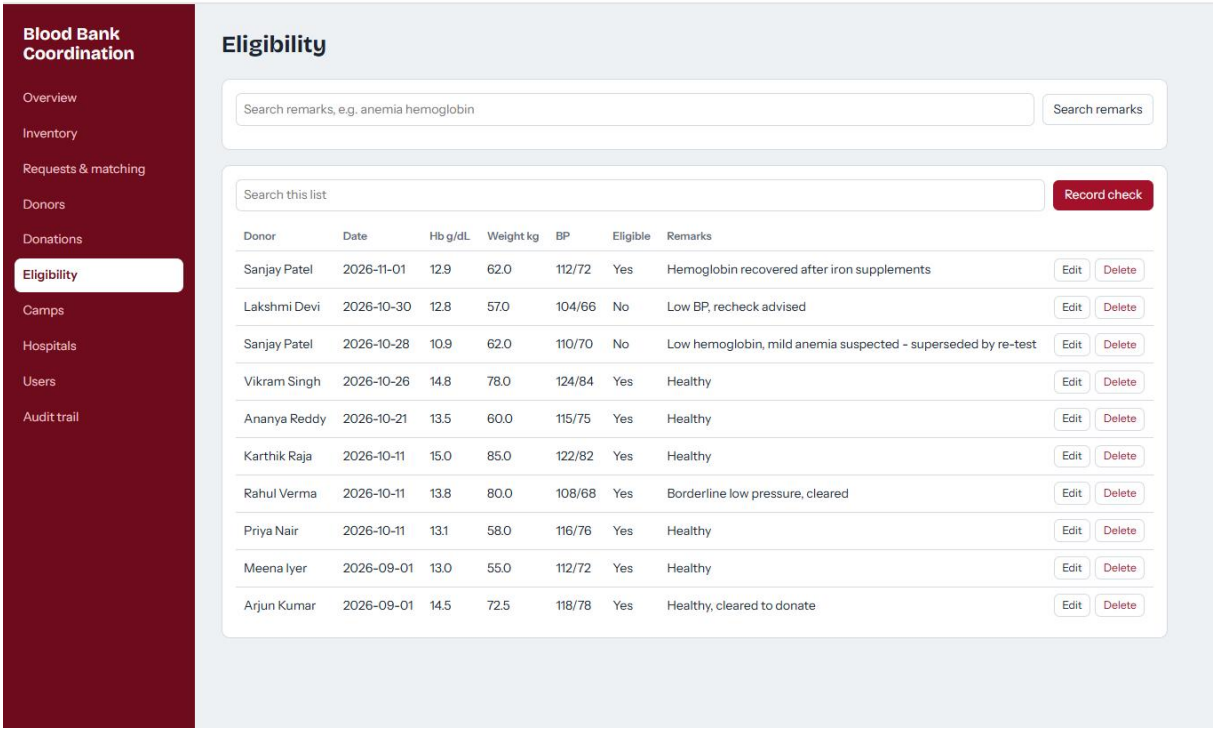


Fig 9.6 Eligibility screen with remarks search

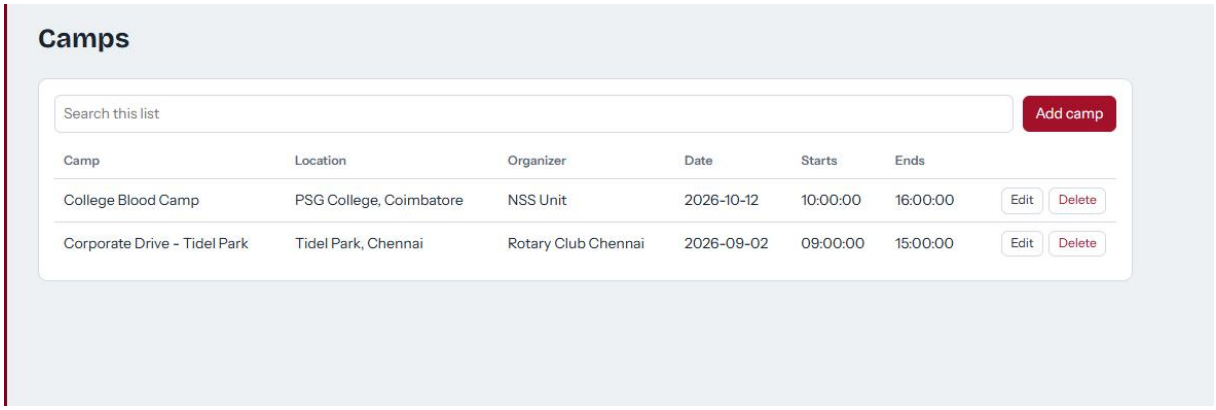


Fig 9.7 Donation camps screen

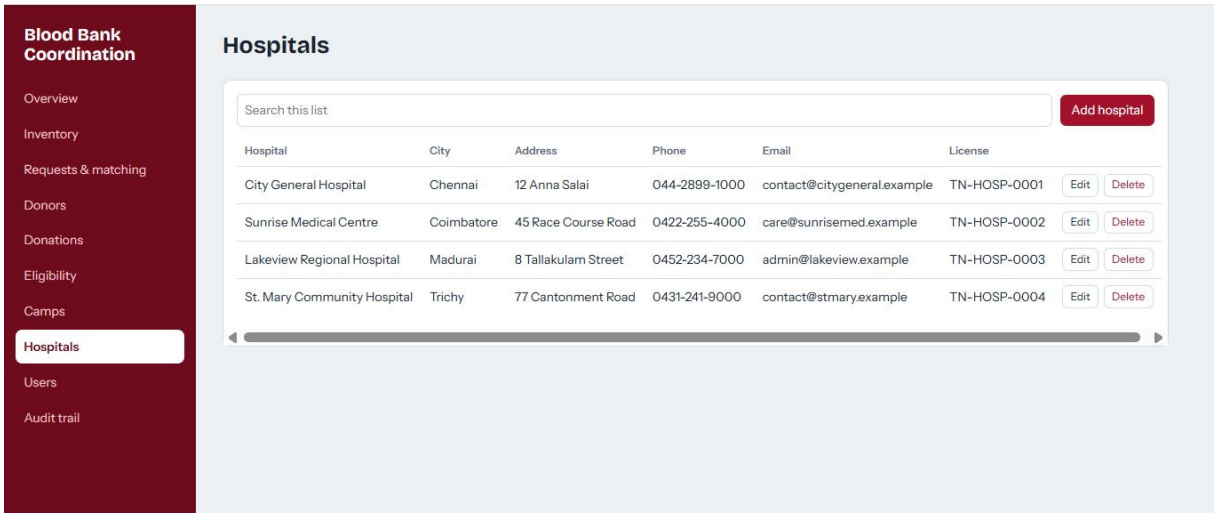


Fig 9.8 Hospitals screen

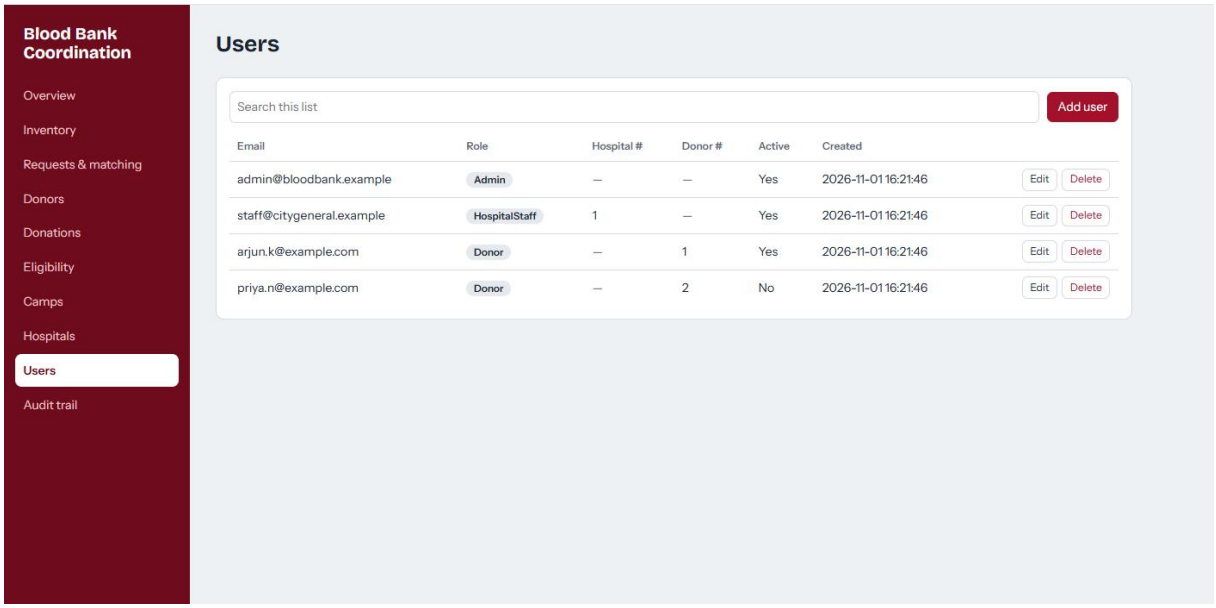


Fig 9.9 Users screen

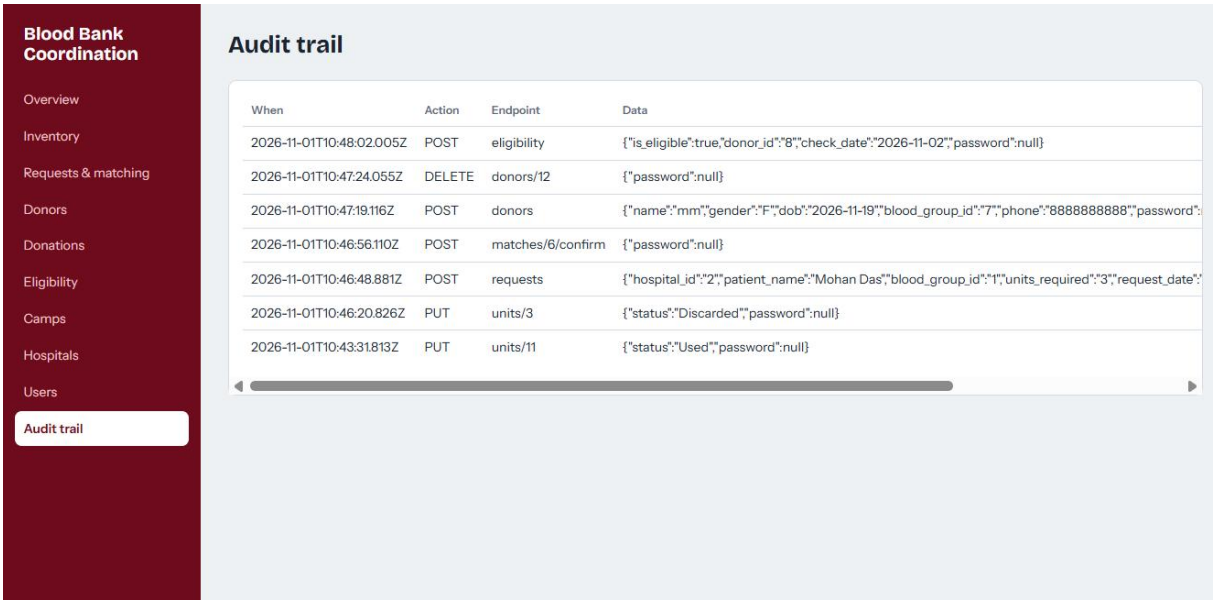


Fig 9.10 Audit trail screen

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[10] J. D. Ullman and J. Widom, *A First Course in Database Systems*, 3rd ed. Upper Saddle River, NJ, USA: Pearson, 2007.

GitHub Link: https://github.com/meghanapillappa/blood_management